
What Moves the Ceiling on Automated Audit: from General Code to Scientific Code, Results and Data

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Abstract

When models write code, report results and prepare data, other models increasingly check them. We study one shipped arrangement for doing so, in which an auditor from a different vendor reads the work and deterministic checks verify what code can verify, and we measure what limits it against ground truth no model supplies. On general-purpose code with hidden test suites, eight readings of the shipped auditor reach 30.0% union recall [20.0, 40.7] at 16.0% flags on correct code, and returns diminish without a plateau. Which model reads, one added rulebook sentence, and the decision rule each move that figure, and on a post hoc rating much of what is missed is behaviour the specification never determined, so a recall figure is a property of auditor, rulebook, rule and benchmark together. We then test the arrangement on science, in three preregistered studies. On reported results, checking that every number traces to its source caught all 49 inconsistencies among report, results and log and none of 25 results that were wrong consistently; the model auditor, recomputing from the code, flagged 23 of those 25. On data, a validator written from the documentation and the auditor caught different faults (92 and 65 of 140) and together 112, at the auditor's 2.9% flag rate on clean data. On scientific code with no tests shown, naming the step rather than the whole problem as the deliverable cut the auditor's flags on correct code from 68.7% to 36.0% on the same programs. Where both tiers ran, the deterministic one flagged no clean item and could not see consistent error; the model saw some of it and erred itself. Every study that made model calls was preregistered, the two unregistered analyses are named, and every report quoted passed independent cross-vendor review.

1. Introduction

When a model writes code, reports a result or prepares a dataset, someone has to check it, and increasingly that someone is another model. CrossAudit, the system every measurement here passes through (§2), is built on two rules for doing that. The model that audits must come from a different vendor than the model that wrote the work, and what a deterministic check can verify is verified by code, not by a model's opinion. An auditor of this kind reads work it did not write and says whether to block it. We ask how much it catches, what decides the limit, and whether the arrangement helps where checking matters most, in science.

A reviewer's score on a benchmark mixes at least four things: whether the model can see a defect, whether it judges the defect worth blocking on, whether the benchmark's hidden tests define the behaviour it would have to see, and how many times it is allowed to look. Work on critic models improves the first two and reports the aggregate gain. We hold the auditor, its instructions and its ground truth fixed and vary the other factors one at a time, against a truth no model supplies: a hidden test suite executed by an interpreter.

The limit on general-purpose code (§3). Eight independent readings of the shipped cross-vendor auditor reach 30.0% union recall [20.0, 40.7] on defective code at 16.0% [10.1, 22.3] on correct code, and the eighth reading still adds 1.9 points, so returns diminish without a plateau. A stronger same-vendor model moves recall by -26.4 points $[-37.3, -15.6]$, a contrast that reverses sign under a weaker severity rule. One added rulebook sentence moves flags by $+26.8$ points [13.6, 40.4] on an exploratory arm without measurably moving outcomes. Of 57 defects no reading flagged, two model raters judge 44 to be behaviour the specification never determined (post hoc). A recall figure is therefore a property of auditor, rulebook, decision rule and benchmark together, and is not portable without its false-positive rate.

Auditing science (§4). We then take the system into science, with one preregistered study for each job a scientist would hand an auditor. On reported results, the deterministic check that every number traces to its source caught all 49 inconsistencies among a report, its results file and its run log, and none of 25 results that were wrong consistently in all three;

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the model auditor, recomputing from the code, flagged 23 of those 25 (21 with a sound finding, post hoc). On data, a validator written from the dataset’s documentation and the auditor caught different faults, 92 and 65 of 140, and together 112, at the auditor’s 2.9% flag rate on clean data. On scientific code with no tests shown, naming the step rather than the whole problem as the deliverable cut the auditor’s flags on correct code from 68.7% to 36.0% and its recall from 87.7% to 72.8%, on the same programs.

Contributions.

- A measurement programme for an automated auditor’s limit against executed ground truth, in which each factor (repeated reading, the reading model, the rulebook, the decision rule, the specification) is varied alone, with union recall always reported beside the false-positive rate it cost.
- Evidence that the limit on general-purpose code is set jointly by the auditor, its rulebook, its decision rule and what the benchmark’s specifications determine, rather than by any one of them.
- The first head-to-head, to our knowledge, of an LLM auditor against deterministic verification on scientific code, reported results and data, with recall and false positives measured against executed or constructed ground truth; the two tiers catch different faults, and each fails where the other does not.
- A methodological finding: in all three science studies the auditor flagged defects in our own construction of items meant to be clean, so an evaluation of an auditor must establish that its clean items are clean by the auditor’s standard, or its false-positive rate measures the evaluator.

Preregistration and review. Every study that made model calls was preregistered before its first one, with hypotheses, intervals and stopping rules fixed in advance, and planned departures recorded as numbered amendments. Two analyses were not registered in advance, and the paper names both (Appendix A). Four preregistered hypotheses were killed by their own rules, and in the science studies one more. Every report was reviewed by a model from a different vendor with access to the raw records, at between three and twenty-one rounds each, and the reviews are archived verbatim. Section 5 discusses what the measurements mean. The full method, results and discussion are in Appendices A–D, and the records, reviews and the command that regenerates every figure are described in Appendix G.

2. The system under study

Every measurement in this paper is taken through one shipped system, CrossAudit. Every auditor prompt is built

by the product’s own prompt builder, and every family but one is read through the product’s own audit call; the exception, `gpt-6-astra`, receives the same prompt bytes through a direct call. We describe it first because its design fixes what the later sections can and cannot vary. We state its structure only here and quote no measurement of it. Its limits are the subject of Section 3 onward.

Separation of roles. A generator model writes an increment (a set of files) into a Git repository. An auditor model then reads the committed bytes and returns findings, each graded **BLOCKER** or **ADVISORY**. The auditor must come from a different vendor than the generator: the audit refuses to run when the configuration names the same vendor for both roles, and when the generator’s vendor is not declared, the receipt records that independence could not be asserted. The same-vendor arms below run that second way, on purpose. The premise is that a model reviewing its own vendor’s output is an interested reader. The product takes that premise as given, and Section 3 puts one consequence of it to the test: whether a same-vendor auditor catches more or less.

Two tiers of evidence. Before the auditor reads anything, a deterministic check layer runs registered checks over the same committed bytes. A check’s finding is *verified*: code produced it and code can reproduce it. A model’s finding is *raised*: it is a reading, not yet reproduced. The checks are grouped into named profiles, so strictness is a setting the project chooses:

- **general**, the default: structured files parse, every local path a file lists as an input or dependency exists, and relative links and leftover placeholders are reported as advisory.
- **science**: results carry a schema; quantities carry units; a result that reports convergence must meet its own threshold; every input a result depends on is listed, exists and is cited at an exact revision (*provenance*); and every number a report states is annotated with the file and quoted span it came from, which a check matches against the cited bytes (*number*→*source*).
- **research**: the general pack plus a citation check under which a report may cite only sources it fetched through a governed tool.

The science profile follows one rule throughout, and the rule is the one Act 4 tests: *the model names its evidence, and code verifies it*. A number with no source line, a unit that disagrees with its schema or an input that does not exist fails without any model’s opinion.

Verdicts are computed, not voted. The verdict is derived by code from the evidence, in a fixed order. A failed deterministic check blocks, whatever the model says. A model **BLOCKER** with no deterministic support either blocks or

escalates to a person, according to a project setting. A blocked increment goes back to the generator for a bounded number of revision rounds, three by default. Each revision passes a repair guard before it is committed: an edit outside the audited scope is refused, and a new catch-all handler, a deleted assertion or a skipped test is flagged to the auditor for the next round.

Receipts. A passing round produces a receipt that binds the verdict to the commit that was audited, the commit of the rulebook the auditor was given, and a digest over every finding with its tier. Every model call is written to a ledger with its prompt hash and cost. A verification command recomputes the digest from the recorded evidence, so a receipt can be checked by someone who does not trust the operator.

What the rest of the paper measures. Acts 2 and 3 (Section 3, with Appendices A and B) measure the model tier on its own. There, an increment counts as flagged when the auditor grades any finding BLOCKER, and the general check pack runs as in the product but decides no recall or false-positive outcome. The model tier is the part of the system whose limit is least known, since a deterministic check catches exactly what it was written to catch. Act 4 (Section 4) sets the two tiers side by side on scientific code, results and data, and asks what each catches that the other does not.

3. The ceiling on general-purpose code

Substrate and ground truth. HumanEval+ and MBPP+ (Liu et al., 2023): Python functions with an executable hidden test suite. One frozen generator (claude-haiku-4-5) was shown each specification and its visible tests and wrote one solution per problem, over two passes of 540 problems. A candidate that passes the visible suite and fails the hidden one is *defective*; one that passes both is *correct*. Recall is measured on all 110 defective instances (56 problems) and false positives on 150 correct ones drawn by seed. The auditor sees the specification, the visible tests and the candidate, never the hidden suite. Because the defective stratum already survived the visible tests, recall here is the fraction of that harder residue caught, not of all defects a model writes (Appendix A).

Estimand and statistics. Union recall at K : the fraction of defective instances blocked (any finding graded BLOCKER) by at least one of K independent readings, averaged exactly over all size- K subsets of the readings. The same union on correct instances is the false-positive rate, always reported beside it. Intervals are 95% problem-cluster percentile bootstraps; paired contrasts carry exact McNemar and cluster sign-flip tests. The shipped cross-vendor route (gpt-5.6-terra) sends no sampling parameter;

same-vendor routes are sent temperature 0, a difference that bounds every cross-family union contrast (§B.2).

Repeated reading. Union recall rises from 10.7% at one reading to 30.0% [20.0, 40.7] at eight, at 16.0% [10.1, 22.3] on correct code (Figure 1). The eighth reading still adds 1.93 points [1.14, 2.78], above the 1.0-point flattening bar registered in advance, so we report diminishing returns, not saturation. Twenty readings pooled across three families reach 48.2% [36.7, 60.0] at 36.0% [28.2, 44.2] on correct code: pooling buys coverage at its own price.

Which model reads. Replacing the auditor with a stronger same-vendor model moved union recall at eight readings by -26.4 points $[-37.3, -15.6]$ (exact McNemar $p = 4.2 \times 10^{-7}$), a preregistered primary. It does not measure auditing ability. The stronger model flags 3.3% of correct code against 16.0%; under an exploratory any-finding rule the sign reverses (59.1% against 34.5%; Figure 3); and at one reading, before the differently sampled unions grow, the gap is -10.1 points. A high-reasoning cross-vendor model reaches 32.7% at 10.7% with four readings, though no paired contrast establishes both advantages together.

The rulebook. One added sentence, *find what the visible tests do not cover*, moved flags on a 56-instance exploratory population from 10 to 25 of 56, $+26.8$ points [13.6, 40.4] ($p = 0.0003$), at $+12.5$ points on correct code, an interval $[-3.6, +26.8]$ that does not exclude zero. On the contrast that isolates the rule, hidden-suite pass rate after one revision moves $+5.4$ points $[-0.9, +12.1]$: the sentence changes what the auditor writes, and whether that makes code pass is not established.

What is missed. Fifty-seven of the 110 defective instances were flagged by no reading of the first three families. Two model raters, one of them the agent that ran this programme, agree post hoc that 44 of 57 (77.2%, [62.1, 91.1]) are cases where the specification’s prose does not settle the value the hidden suite expects, reversing an earlier classification of the same instances as unexercised edges. Reading six families at forty readings shrinks the never-flagged set to 32 of 110, and 43 of the 68 instances both raters call undetermined are flagged by some reading, so being undetermined is not sufficient for being missed; the undetermined share of what remains is stable (25 of 32). Where an outside paper had independently named twelve MBPP specifications as defective, our raters agree on 9 of the 11 that overlap. An attempt to construct a population whose specification-determinedness would hold by construction failed its own review, so this remains a post hoc description (Appendix B.5).

What follows. A union recall figure is contingent on the false-positive rate it cost, the model and sampling of the route, the rulebook, the decision rule, and what the benchmark’s specifications determine. Eight further studies, re-

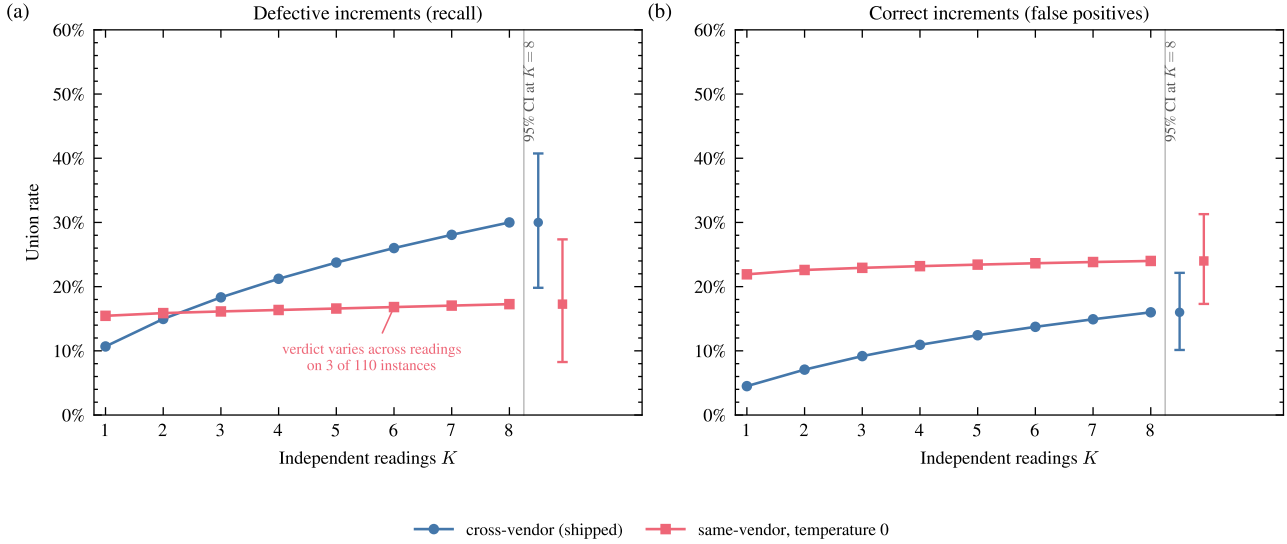


Figure 1. Returns to repeated reading diminish short of complete. Union rate against the number of independent readings K , for two auditor families. (a) Recall on defective increments; (b) false positives on correct increments. Bars at the right of each panel give the 95% problem-cluster bootstrap interval of each series at $K = 8$, dodged in x so overlapping intervals stay separable. The cross-vendor curve does not meet the preregistered flattening bar (its last-step gain is 1.93 points [1.14, 2.78] against a bar of 1.0) and is not described as saturating. The same-vendor curve meets it, at 0.23 points, but that route is sent temperature 0 and its verdict varies across readings on only 3 of 110 defective instances under this configuration, so its flatness is not evidence of exhausted headroom; whether it reflects the sampling or the model is not separated here (§B.2).

ported in Appendix B.5 with what each licenses, include a specification-rewriting intervention that raised adjudicated diagnosis only where the added rule agreed with the hidden suite, and a second substrate that closed without a quotable number.

4. Auditing AI for science

A scientist would hand an auditor three jobs: check the code that computes a result, check that a reported result follows from that code, and check the data an analysis starts from. We ran one preregistered study for each, with ground truth no model supplies: scientists’ numeric tests for code, re-execution for results, and injected faults for data. Every item is audited through the product’s own audit call and shipped rulebook, with the same flag rule as above. Construction, gates, amendments and full results are in Appendix C.

4.1. Scientific code

Design. SciCode (Tian et al., 2024) decomposes research problems from 16 subfields into steps with numeric tests written by scientists. The same frozen generator wrote every step three times, each sample building on its own earlier steps. An instance is correct if it passes the step’s tests and defective if not; a step enters only if some sample passes it. This gives 81 defective instances (30 problems) and 150 correct ones (53 problems). SciCode has no visible/hidden

split, so the auditor sees the step’s prose, background, header and code, and no tests. The shipped cross-vendor auditor and the generator’s own model each read every instance eight times. After the registered arm’s outcomes, a second registered arm changed only the task text, naming the step’s function as the deliverable instead of opening with the whole problem; the program bytes are identical (Appendix C.1).

Results. In these runs, what the auditor was told the deliverable is changed its flag rate substantially (Figure 2a). With the whole problem as the task, the shipped auditor flagged 87.7% [72.7, 98.6] of defective and 68.7% [56.6, 79.7] of correct instances at eight readings, many of its objections to correct code being that one step did not implement the whole problem. With the step as the deliverable, flags on correct code fell to 36.0% [26.4, 45.6], a paired change of -32.7 points [$-42.8, -22.4$], and recall fell to 72.8% [59.3, 87.3], -14.8 points [$-28.6, -2.7$]. The generator’s own model flagged 46.9 points less of the defective instances than the shipped auditor in each arm (a route comparison). At one reading the shipped auditor flags 61.7% of defective against 19.8% of correct instances: it discriminates, modestly. A read sample of its flags on correct code ranges from real divergences between code and step text to contradictions inside the benchmark’s own instructions and tests too weak to separate outputs. Few defects escaped all sixteen readings (7 and 19 in the two arms); most were rated consensus-undetermined by the prose, but the registered test that this share exceeds the general-code residual’s 77.2% is

Table 1. Every primary quantity this paper quotes, with the population it is measured on and the label its source report gives it. Values are percentages or percentage points. Intervals are 95% problem-cluster percentile bootstraps unless the label names another method. The coverage simulation of §A.3 measures single-rate intervals only; the contrasts and the category-conditioned shares have no measured coverage. The two third-rater rows rest on labels from a rater that agreed with itself on 7 of 11 repeated items (6 problems). The correct-code row counts disagreement on inputs extracted from the findings; of the 20 flagged instances, 9 agreed on every extracted input and 9 yielded no valid input, which can be neither confirmed nor refuted. Every row is read from a committed record by `figures/make_table1.py`.

quantity	population	value	interval	label
Union recall, shipped route, $K = 8$	110 defect, 56 problems	30.0%	[20.0, 40.7]	preregistered
Union false positives, shipped, $K = 8$	150 correct, 143 problems	16.0%	[10.1, 22.3]	preregistered
Pooled union, 3 routes	110 defect	48.2%	[36.7, 60.0]	preregistered
its false positives	150 correct	36.0%	[28.2, 44.2]	derived (C1)
Gain from the eighth reading	110 defect	+1.93	[1.14, 2.78]	bar 1.0, not met
Same-vendor stronger – cross, recall	110 defect	–26.4	[–37.3, –15.6]	preregistered
its false-positive difference	150 correct	–12.7	[–19.3, –6.1]	prereg.; 3.3% v 16.0%
Rulebook rule – shipped, recall	110 defect	+30.9	[+19.1, +43.1]	preregistered
its false-positive cost	150 correct	+20.7	[+12.8, +28.8]	preregistered
Rulebook rule, flags	56 loop defect	+26.8	[+13.6, +40.4]	exploratory
its cost on correct code	56 loop correct	+12.5	[–3.6, +26.8]	unconditional, approx.
isolating contrast, pass rate	112 loop, 96 problems	+5.4	[–0.9, +12.1]	registered, failed
<i>astra</i> – cross, common $K = 4$	110 defect, 56 problems	+11.5	[+2.9, +21.3]	exploratory
its false-positive difference	150 correct	–0.3	[–3.8, +3.4]	exploratory
Undetermined, sheet groups	121 entries, 110 instances	+19.8	[+1.2, +38.1]	registered method
one definition of caught	57 against 53	+23.3	[+2.0, +43.6]	post hoc
Undetermined share, 3-family residual	44 of 57	77.2%	[62.1, 91.1]	post hoc
6-family residual	25 of 32	78.1%	—	post hoc (C14)
Correct-code flags, reference disagrees	2 of 20 flagged	10.0%	[0.0, 26.3]	preregistered (C16)
Clarified – original, diagnosis	32 selected, 19 problems	+28.1	[+7.7, +51.4]	one-signed; exact $p .0625$

killed in both arms by its own interval rule.

4.2. Scientific results

Design. Each of 74 correct SciCode instances (36 problems) whose output on one seeded test case is a real scalar or a short real array becomes a results increment: the program, the test’s inputs, a script that runs one on the other and writes a run log, a `results.json` citing each value’s log line, and a report whose numbers carry the product’s number→source annotations. Each yields a clean item and a faulty one. Five fault types are *reporting* faults, where the artefacts disagree with each other or with the log; two are *fabrications*, a $\times 10^3$ slip or a 5% error written consistently into log, results and report. The LLM auditor reads each item four times with the shipped *science* profile on, the profile’s own verdict comes from the same call, and re-execution compares the output with the report (Appendix C.2).

Results. The deterministic profile flags all 49 reporting faults and none of the 25 fabrications, because in those everything agrees with everything else (Figure 2b). The LLM auditor flags 23 of the 25 fabrications, 21 with a sound finding about the fault (post hoc), usually by working the result out from the code, for example evaluating a KL divergence to 1 where the report says 1000; the other two rest on wrong calculations. All 23 discordant items go the

LLM’s way (exact McNemar $p = 2.4 \times 10^{-7}$). On reporting faults its rate is of the combined workflow, since its prompt carried the profile’s findings. Re-execution misses four faults on outputs within `np.allclose`’s default absolute tolerance of zero. The LLM also flags 12 of 74 clean items: 8 correctly identify defects in our construction (a provenance sentence false where the benchmark passes inputs as call literals; a unit label contradicting documented units), 4 rest on wrong calculations or a syntax error that is not there.

4.3. Scientific data

Design. Four CC BY 4.0 physical-science tables (Yeh, 1998; Brooks et al., 1989; Tüfekci, 2014; Hamidieh, 2018); 280 items of 120 rows with a data card giving each column’s meaning and unit: 140 clean, and 140 carrying one of seven faults (mixed units, negated values, duplicated rows, a shuffled target, swapped columns, a –999 sentinel, rounding). Both model families read each item four times, against a validator written from the card and frozen before any item existed (Appendix C.3).

Results. The two tiers catch different faults (Figure 2c). The validator flags 92 of 140 faulty items (65.7% [57.9, 73.6]) and no clean item; the shipped auditor 65 of 140 (46.4% [38.6, 54.3]) at 4 of 140 clean items (2.9%); their union 112 (80.0% [73.6, 86.4]) at the same 4. Of the 20 faulty items the auditor flags and the validator misses, 18 carry

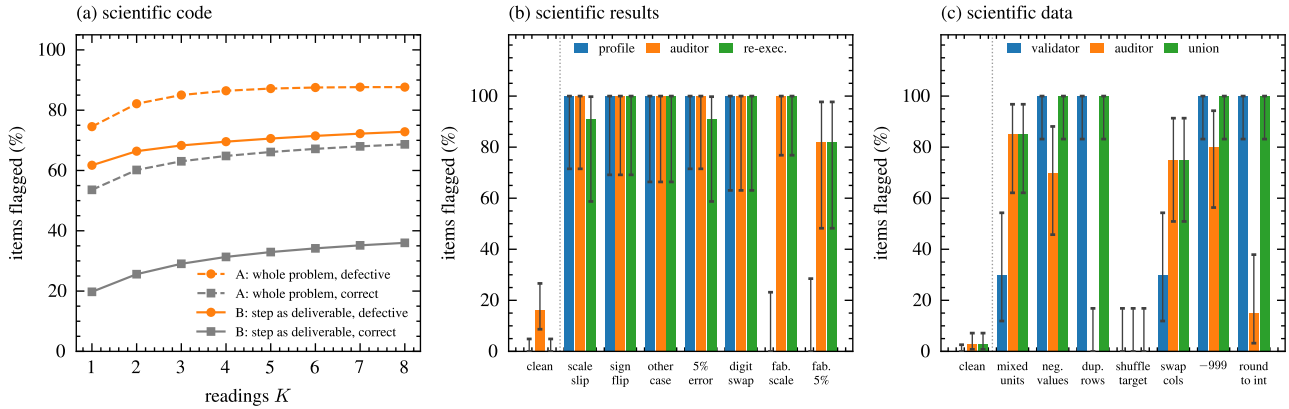


Figure 2. What each tier flags. (a) Scientific code: the shipped auditor’s union flag rate against the number of readings, on defective (orange) and correct (grey) instances, in the registered arm that gave the whole problem as the task (dashed) and the arm that named the step as the deliverable (solid). (b) Scientific results and (c) scientific data, by fault type, with clean items leftmost: the deterministic tier (blue: the *science* profile in b, the card-derived validator in c), the LLM auditor at four readings (orange), and re-execution (b) or the union of the two (c) in green. Bars carry exact binomial 95% intervals on the item counts (supplementary; they ignore clustering). Registered flag rule throughout: any model BLOCKER in the union of readings.

a finding about the injected fault itself (post hoc), mixed units or swapped columns; in one it argued that three atoms spanning a 135 pm radius range cannot have a mean radius of 1.6 pm, which is the converted cell. It flagged no shuffled target and no duplicated rows (0 of 20 each), and did worse than the validator on every fault the validator was written for. The generator’s own model flags 67 of 140 clean items, and for at least 48 of those one finding states a row count, field count, missing column or truncation the file contradicts. The validator omits one range the concrete documentation states and uses a list of continuous columns the card does not give.

5. Discussion

A recall number is an operating point. Recall and false positives move together with any severity threshold, and a critic can be tuned to any recall by blocking more. On general-purpose code the same auditor sits at different points under different decision rules, repetition buys recall at a roughly constant price in false positives, and pooling families buys coverage at a higher one. On scientific code the operating point also moved with what the task named as the deliverable. A recall figure quoted without its false-positive rate, its rulebook, its decision rule and its task framing does not carry to another setting.

What the science studies add. They give the science profile’s design rule (the model names evidence, code verifies it) an empirical reading and a boundary. On results, number→source verified every inconsistency between a report, its results file and its log, and could not see a result that was wrong consistently; the model, by recomputing, flagged 23 of 25 such fabrications. On data, a validator written from

the card caught what a rule can state and missed what needs reading across columns and units, which the model caught in 18 of the 28 such items the validator missed. Where both tiers ran, the deterministic one flagged no clean item and the model carried the errors; on data the union was better than either part, and on results the model’s flags already contained the profile’s, whose findings its prompt carried. Neither study shows that the model tier scales: the artefacts were small, the model’s fabrication findings are hand calculations (several wrong), and it missed every fault that needs comparison across many rows. A model reading is a useful second tier where the artefact fits in the reader’s working memory, not a substitute for executing the code or computing over the data.

Clean items must be clean by the auditor’s standard.

In all three science studies the auditor flagged defects in our own construction: a card and file that named a column differently, a file outside the audited scope, a report claiming a run no file performed, a provenance sentence false for inputs passed as literals, a unit label contradicting the code’s documentation, and a task that named the whole problem when one step was delivered. Each was a correct flag on an item we had built as clean. We caught two in pilots, one when every verdict came back as an escalation, and the rest only by reading every flag on a clean item, which the registrations required. Without that reading, the false-positive rate would have measured the evaluator.

Limits. The auditor families are not sampled alike, which bounds every cross-family union contrast, and no family was compared at a matched false-positive rate. Primary intervals are problem-cluster bootstraps whose coverage was simulated only for single rates. Every rating was made

by a model, one of them the agent that designed the studies; no human rated anything. The reviewer of every report is also one of the measured families. The science studies use one benchmark or four tables each, synthetic faults, small artefacts and one generator. The full limits, and the experiments that would remove them, are in Appendices D and E.

6. Related work

Critics and repeated reading. Trained critic models catch more bugs in model-written code than unaided humans and hallucinate some (McAleese et al., 2024); aggregating several review passes raises review F1 on real pull requests (Zeng et al., 2025); and coverage under repeated sampling rises steeply and then flattens (Brown et al., 2024). The natural-defect setting, the precision cost of a louder critic and the shape of the repeated-sampling curve are therefore known. What we add is narrow on this axis: the union of independent readings as an instrument for locating a limit, ground truth supplied by an interpreter, and the false-positive rate beside every recall figure.

Who audits whom. Self-correction without external feedback does not reliably improve reasoning (Huang et al., 2024), and model judges prefer their own generations, harmfully on instances they get wrong (Chen et al., 2025). Our same-vendor arms are the code-audit case of that literature; because routes differ in model, operating point and sampling, we report their contrasts as contrasts between routes, not as self-preference.

What the hidden suite decides. EvalPlus rebuilt HumanEval and MBPP with larger suites (Liu et al., 2023); models collapse onto one reading of an underspecified task, and twelve MBPP tasks were named as such by hand (Richter & Papadakis, 2026). Generated tests err mostly in the expected value (Xiong et al., 2023), and generated oracles tend to capture what a program does rather than what was intended (Konstantinou et al., 2024). Our claim is the auditing counterpart: where the prose does not fix the value, a recall number computed against that suite measures the benchmark as much as the auditor. Consensus filters for generated tests (Taherkhani et al., 2026; Xu et al., 2025) and self-consistent errors (Tan et al., 2025) bear on one of our appendix studies (Appendix F).

Auditing scientific work. On manuscripts, no model exceeds 21.1% recall or 6.1% precision on errata-grade errors and repeated runs rarely rediscover the same errors (Son et al., 2025); the best models detect 46.7% of real paper-code discrepancies (Baumgärtner & Gurevych, 2026). An LLM auditor of AI-scientist projects detects hand-labelled pitfalls far better with logs and code than with the paper alone (Luo et al., 2025); pattern-based linting targets

methodology bugs in scientific Python (Samsonau, 2026); cleaning agents find row-level errors in corrupted tables and miss errors spanning rows (Bendinelli et al., 2025); and simulation code can run and still solve the wrong equations (Song et al., 2026). SciCode supplies scientist-written tests (Tian et al., 2024), and reproduction benchmarks ask agents to recover reported results (Siegel et al., 2024). We found no prior measurement of an LLM auditor’s recall and false-positive rate against executed or constructed ground truth on scientific code, results and data, set beside the deterministic checks a shipped system runs on the same artefacts.

Preregistration of evaluation protocols is beginning to appear in machine learning (Vaccaro, 2026; Thomas et al., 2026); we count our use of it as method, not contribution.

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A. Method in full

A.1. Substrate, population and ground truth

Every result in this paper is measured on one substrate: HumanEval+ and MBPP+ (EvalPlus), Python functions whose median task is 41 words of prose and 6 lines of reference code, each with an executable hidden test suite. Every stratum, recall denominator and false-positive population is therefore decided by an interpreter, and no model judges a recall or false-positive outcome. The one outcome defined by adjudication, correct diagnosis in the clarification study, is judged jointly by two models and is labelled so where it appears. The benchmark’s base suite is shown to the generator as the *visible* suite, and its plus suite is held back as the *hidden* one. A second substrate, the standard-library half of BigCodeBench, was built and audited and closed without a quotable number when its preregistered criterion was not met on a third run; nothing from it is reported here.

How the population was built. The corpus has 542 problems. Two, HumanEval/32 and Mbpp/590, were excluded by name before generation because their own reference solutions fail their own hidden suites in this harness, which leaves 540. One frozen generator, `claude-haiku-4-5-20251001`, was shown each specification and its visible suite and wrote one solution per problem. A candidate that passes the visible suite and fails the hidden one is a *defective* instance; one that passes both we call *correct*; one that fails the visible suite is discarded from every recall and false-positive population. The defect stratum is small, about 10% of solutions, so under a rule registered before the second pass and depending only on execution outcomes, generation was repeated over the same 540 problems until there were at least 100 defective instances or three passes. It stopped at two: 56 and 54 defective, 455 and 455 correct, 29 and 31 discarded. An instance is therefore a (pass, problem) pair, and one problem can contribute two. Recall is measured on all **110** defective instances, which come from **56 problems**; false positives on **150** of the 910 correct instances, drawn once from a fixed seed before any auditor ran, from 143 problems. Seven of the 110 defective instances fail by not terminating rather than by a failed assertion, and the registered population keeps them.

Two consequences of these definitions hold throughout. *Correct* means *passes the hidden suite*. It is a benchmark label produced by an interpreter, not a guarantee that the program meets its specification, and a suite can be incomplete or wrong. Since this paper’s own third result is that a specification often fails to determine what the hidden suite asks, we cannot treat passing that suite as correctness in the ordinary sense, and a *false positive* is likewise a flag on test-passing code rather than a demonstrated mistake by the auditor. And the defect population is not a sample of model-written code: it is the residue that already survived the visible tests, which is by construction the harder part. A recall figure here is the fraction of that residue an auditor catches, never the fraction of all defects a model writes.

A.2. The auditor and the estimand

The auditor reads the specification, the visible tests and the candidate, and returns findings with severities; an increment is blocked when any finding is graded BLOCKER. It never sees the hidden suite. The estimand throughout is **union recall at K** : the fraction of defective instances blocked by at least one of K independent readings, averaged exactly over all $\binom{K_{\max}}{K}$ subsets rather than by sampling. The matching quantity on the correct stratum is the union false-positive rate, and a union recall figure is reported only beside the false-positive rate it cost.

Four auditor routes appear below. The shipped cross-vendor auditor is `gpt-5.6-terra`; the same-vendor auditor is `claude-haiku-4-5`, the generator’s own family and size; the stronger same-vendor auditor is `claude-sonnet-4.6`; and a third family, `gpt-6-astra` at high reasoning, is read at $K = 4$. The auditor’s constitution, its prompt builder and every route’s capability card are in the auditor harness, a separate repository, at the commit named in the provenance file, whose per-reading records carry the hash of every prompt actually sent.

The routes are **not sampled alike**, and this bounds every cross-family union-of- K contrast in the paper. The cross-vendor route’s capability card does not permit a sampling temperature, so no sampling parameter is sent and the route varies between readings. The same-vendor routes permit one, and the layer sends temperature 0, so those routes are nearly deterministic. A union over readings that do not differ cannot grow, so the two families’ unions are not equally free to rise with K , and a cross-family difference in union recall confounds auditing ability with how much each union could grow. The asymmetry is a property of the shipped routing layer we measured rather than a design choice. We did not discover it until after the runs, and we report the $K = 1$ contrast beside the $K = 8$ contrast wherever it matters. A temperature-matched replication is the first item in §D.6.

A.3. Statistics

The unit of observation is the instance and the resampling unit is the problem, because problems recur across generation batches and instances sharing a problem are not independent. The K readings of an instance are repeated measurements of it, not K further independent observations. The bootstrap over problems estimates uncertainty from which tasks the benchmark happens to contain. It does not estimate how much a fresh batch of model calls on the same tasks would move the numbers, which would need replicate runs we did not budget for at every arm.

Every rate carries a 95% percentile bootstrap over whole problem clusters (10,000 resamples, seeds fixed in each preregistration), with the Wilson interval printed beside it and marked as too narrow. Paired contrasts carry an exact McNemar test and a cluster-level sign-flip permutation test. Where every discordant pair points one way, the percentile bootstrap's bound at zero is an artefact, and an unconditional interval (grid-approximated, unclustered) is quoted instead. The one exception is the clarification study, whose registration fixed the bootstrap for a contrast that turned out one-signed; its interval is reported as registered and read beside its exact cluster test (§B.5). A saturation curve is summarised by a constrained exponential fit, and the fitted asymptote is quoted only when the preregistered flattening bar is met; otherwise it is labelled an extrapolation and the raw union at $K_{\max}$ is the number quoted. Counts of five or fewer are quoted as counts. Multiplicity is handled per study as each registered it: the study carrying the primary contrasts corrects over every comparison it computed, a Bonferroni family of sixteen with a threshold of 0.00313 that two comparisons clear; others registered a single primary, and study 19 reports its twenty further quantities without correction.

We measured interval coverage by simulation rather than assuming it. Under the two strata's actual cluster profiles (110 instances in 56 problems, and 150 in 143), with within-problem concordance 0.5, the percentile bootstrap's coverage of a nominal 95% rate interval is as follows, over 400 replicates of 600 resamples each. The Monte Carlo standard error is about 0.011 near 95% coverage and 0.020 at the lowest cell; the simulation covers single rates only, and uses integer-index percentiles where the analyses interpolate (`analysis/coverage_simulation.py`).

true rate	0.03	0.05	0.10	0.16	0.30	0.48	0.60
defect (56)	0.798	0.870	0.933	0.933	0.953	0.935	0.928
correct (143)	0.888	0.918	0.938	0.933	0.943	0.955	0.965

In these simulations coverage is near nominal at the true rates this paper's headlines sit at and **badly anticonservative at small true rates**: at a true rate of 0.03 on the defect stratum the simulated interval covers about 0.80, not 0.95. This is evidence of potential undercoverage for analogous intervals, not a measured coverage of any interval quoted here. Intervals on rates below roughly 0.10 may be narrower than they claim. The simulation costs no model calls. Two earlier paired-contrast constructions in this programme were found by review to cover 0.416 and 0.075 where 0.95 was claimed, and both were replaced. No sample size was justified in advance by a minimum-detectable-effect calculation.

An interval that contains zero is reported as one that does not exclude zero, and nothing more. No equivalence margin was registered anywhere in this programme and no equivalence test was performed, so none of those intervals is evidence that two conditions are alike or that an intervention did nothing. This work once inferred equal cost from an interval containing zero and withdrew the inference on review. Where a hypothesis fails its registered criterion we say it failed, not that the effect is absent.

All analysis is plain Python 3.13 using only the standard library (`random`, `math`, `itertools`, `csv` and `json`). The bootstrap, the exact McNemar test, the cluster sign-flip permutation and the exact subset enumeration behind every union-at- K figure are implemented in the repository rather than called from a numerical package, so each can be read where it is used and none depends on a third-party version.

A.4. Preregistration and independent review

Who did what. The studies were designed, run, analysed and written up by an AI research agent, Claude (Anthropic; `claude-opus-5` and `claude-fable-5-1`, working in Claude Code), under the direction of the author, who is responsible for the work. Wherever a rating or adjudication below is attributed to L1, the rater was that agent. It wrote the rubrics, knew the hypotheses and the earlier labels, and is from the same vendor as the generator. **No human rated or adjudicated anything in this paper.** Every report was then reviewed by a second model, `gpt-6-astra`, as described below.

Each study that made model calls was preregistered before its first one, fixing its population, hypotheses, primary outcome, intervals, kill conditions, budget and stopping rule, with two exceptions. First, the severity sweep of §B.2 re-grades findings already archived, makes no model calls, and was never registered: every figure it produces is post hoc and exploratory, and none is a confirmatory test. Second, the analysis behind §B.5's third-rating result was registered a day *after* its contrast had been computed and committed, in a document that claimed the opposite; that document is withdrawn in the repository and the analysis is registered nowhere. Its method was fixed in advance by an earlier amendment, but the disjoint reading, the abstention-excluded secondary and the broken-sheet comparison were not.

Planned departures are numbered amendments committed before the step they govern, each recording what was known at the time. Deviations discovered only afterwards, such as a leak check or a spend guard implemented more narrowly than registered, are disclosed in the report and are not amendments. Where an amendment followed an interim observation, the results say so and argue whether the change could favour the hypothesis.

Each report was then reviewed by a model from a different vendor to the generator, with read access to the raw records, asked to re-run the analysis, recompute the intervals independently, and check every sentence against the records. **That reviewer is gpt-6-ast-ra**, which is also one of the auditor families measured in §B.2, one of the two raters behind §B.4's labels, and one of the two adjudicators whose joint labels define the clarification study's outcome. For most claims this is a cross-vendor check on work the reviewer had no part in. For the claims that rank *ast-ra* against the other families, and for the clarification study's adjudication, the reviewer is the subject, and those results should be read as self-assessed until a model that audited nothing in this programme has checked them. This project's own P3 Amendment 10 disqualified the same model as the clarification study's manipulation-check rater for the same reason, yet kept it as that study's second adjudicator and its reviewer; the standard was applied unevenly within one study, and not applied to the review at all until a pre-submission referee pointed it out.

Reviews are archived verbatim, including those that refused approval, and several studies needed many rounds. On one day five reports were refused at once, and clearing those five took thirty-one further rounds. In most of them what moved was what the reports claimed about their measurements, and in most of the later refusals the object was a repair written to satisfy the previous round rather than the study it repaired. A fix written to satisfy a reviewer is itself an unreviewed claim; that is the most transferable thing this process taught us.

The reviewer is a language model, and this paper is about the ceiling on language models reading code they did not write. That applies to our own instrument. The studies of §B.5 cleared review at rounds 3, 3, 4, 6, 7, 9, 10 and 11, and the ceiling study behind the headlines at 21; in every case the round before the last still returned substantive findings. Some of those findings were defects introduced by the repairs between rounds, so the counts do not estimate a single pass's recall; they show only that no report cleared in fewer than three. *Quotable at round N* means a report survived *N* passes, not that it was verified, and the marks in this paper should be read as a filter of unknown recall.

Two further points follow. First, measurements did move on review. The severity sweep's first review found its loader misreading one family's draws and another's population, and every cross-family figure was recomputed; the clarification study's second review moved its primary from 9 of 31 to 9 of 32 instances and its contrast from +29.0 to +28.1 points, by finding an instance wrongly excluded; and a label count inside a reviewed report was 71 where its data say 67. Once a figure's inputs were right, the reviewers reproduced rates, intervals, subset enumerations and seeds independently and found no arithmetic error, and most refusals were about what the prose claimed. On recomputation the instrument is strong; on interpretation it is weak, and the two should not be quoted as one assurance. Second, the class of defect this leaves open is the one that looks right to a language model. The clarification study audited the wrong program for 384 readings, because a per-problem field was used where a per-instance one was needed, and every automated gate passed: each checked that the code was *identical across conditions*, and it was, identically wrong. The agent found it while reading an adjudication sheet, by noticing that a displayed program returned exactly what its hidden suite was said to reject. Neither the gates nor a reviewer found it, and we do not know what else of that kind remains.

Every review prompt in this programme stated the claimed result before the reviewer began. We tried to test what that costs, and the attempt failed. One report, quotable at round 6 after five refusals and unchanged since, was dispatched again under a prompt whose own text withheld every figure and every prior verdict and asked for an independent computation first. That prompt's list of binding artefacts then named two files that state those figures, one of them the claims-ledger entry for the result under review. The blinding was not enforced, so the agreement between that pass and the briefed one establishes nothing. What the pass did return is a disagreement count: six substantive findings that six briefed rounds on the same file had not produced, among them that the report's two groups were two different definitions of *caught* rather than one

population with duplicate rows, that its rating sheet displays at most three failing inputs while 81 of its 121 entries have more recorded, and that a count of changed labels was 71 where the data say 67. **That count is $n = 1$, uncontrolled and unregistered.** A seventh round might have found new things under any prompt, and no matched briefed round was run against the same file. We report it as a reason to test blinding properly rather than as a result about blinding. Later first rounds withheld the conclusions from the prompt text, but the report under review stayed among the named artefacts, with an instruction to derive the result before reading it; that is a request, not blinding.

Every study drawn on here has completed that review. One further study was re-run twice and closed without a quotable number; the paper names it only to say what it no longer supports.

B. Results on general-purpose code in full

The claims this paper defends, each with its interval and its label, are listed in `CLAIMS.md`. A claim enters that file only after its report has passed an independent review by a model from a different vendor with access to the raw records, and every section below draws only on such reports. Section B.5 reports eight studies whose reviews took between three and eleven rounds each to reach that point, and states what each does and does not license. The headline results of §B.1–§B.4 come from the ceiling study; where they cite a study from §B.5, they say so.

Rates carry the 95% problem-cluster percentile bootstrap as the primary interval, with the Wilson interval beside it where the source report gives one, because problems recur across generation batches and an interval that treats instances as independent is too narrow. For single rates, simulations of an analogous interval under this clustered design (§A.3) suggest coverage near nominal at the headlines’ rates and possible undercoverage below a true rate of about 0.10, under an assumed within-problem correlation. They do not measure the coverage of the intervals reported here, and the paired contrasts and category-conditioned shares have no measured coverage.

Sixteen comparisons were computed in the study that carries the primary contrasts, under one Bonferroni family with a threshold of 0.00313. That correction is applied to p values and never to intervals. Every interval in this paper is an unadjusted 95% interval, so a directed statement resting on an interval alone, with no p value that clears its family, is worded as a direction rather than as a finding. Each p value belongs to the family of the study that computed it. Those in §B.3 are members of that sixteen-member family. The one in §B.2 is study 18’s single registered primary, which had no family to correct within; it would also clear 0.00313. The clarification study’s are reported beside a primary registered on its interval, and no correction was registered for them. Where a p value appears below, the exact McNemar value and the cluster-level sign-flip permutation value are both given, because instances within a problem are not independent and only the second respects that.

Table 1 collects every primary quantity below with its population and its label; the prose gives each one where it is used.

B.1. Returns to repeated reading diminish short of complete

The shipped cross-vendor auditor reads an increment holistically and returns findings; an increment is blocked if any finding is graded BLOCKER. Over eight independent readings of the same 110 defective instances, union recall rises from 10.7% at one reading to 30.0% [20.0, 40.7] at eight (Figure 1). The eighth reading still adds 1.93 points, cluster [1.14, 2.78], against the flattening bar of 1.0 points registered before the first model call, so the bar was not met. Under exact subset averaging that statistic has a closed form: the number of instances flagged by exactly one of the K readings divided by Kn , here $17/(8 \cdot 110)$. It is a count of singletons, not a fitted slope. Its interval resamples problems and so carries task-sampling variance, while the statistic itself is made of reading-sampling variance, which the interval does not describe; the bar should be read as a statement about these 17 instances rather than about a population slope.

The constrained fit puts the asymptote at 31.5%, but on our own registered criterion that figure is an extrapolation, and the raw union at eight readings is the number to quote. What the data support is diminishing returns at the budget we could afford, not saturation: at 1.9 points per reading the curve was still climbing when we stopped. The bar also depends on where a family was stopped. Exact subset-averaged increments do not grow with K (this curve’s is 2.9 points at the fourth reading and 1.9 at the eighth), so a fixed bar is harder to meet at $K = 4$ than at $K = 8$, and meeting it at either says little by itself about where the curve ends.

Twenty readings pooled across three auditor families reach 48.2% [36.7, 60.0], and the same twenty readings flag 54 of 150 correct increments, **36.0%** [28.2, 44.2]. That is more than twice the shipped route’s 16.0%, so pooling buys its coverage at its

own price. Those twenty readings are also not twenty opportunities. The pool is eight cross-vendor, eight same-vendor and four *astra* readings, and the same-vendor route gains 1.8 points from its first reading to its eighth against the cross-vendor route's 19.3. It is sent temperature 0 and returns nearly the same verdict each time, though this design cannot say how much of that is the temperature and how much the model. Counting that route as roughly one reading puts the depth nearer thirteen than twenty (a heuristic, not an estimate), and the pooled figure should be read as a union over three routes rather than over twenty independent looks.

The price is measured on the same footing. Union false positives on 150 correct increments reach 16.0% [10.1, 22.3] at eight readings. The recall bought per point of false positive, meaning recall gained over one reading divided by false positives gained over one reading, is **1.68** at eight, and it does not worsen as readings accumulate: from $K = 2$ to $K = 8$ the ratio runs 1.67, 1.63, 1.64, 1.65, 1.66, 1.67, 1.68. It falls once and then rises, so repeated reading on this route buys recall at a slightly improving price. The ratio is a quotient of two measured rates and carries no interval of its own.

B.2. Which model reads, and under which severity rule

Replacing the auditor with a stronger model of the generator's own vendor moved union recall at eight readings by -26.4 points, cluster $[-37.3, -15.6]$: 4 of 110 against 33 of 110. Three instances were flagged only by the new family and thirty-two only by the shipped auditor, at an exact McNemar p of 4.2×10^{-7} and a cluster sign-flip p of 2.0×10^{-5} . This was study 18's single preregistered primary, so no family correction applies. Three measurements qualify what it means, and none of them favours reading it as a statement about auditing ability.

First, the two auditors are not at the same operating point, which is the comparison this paper elsewhere warns against. The stronger model flags correct code at 3.3%, or 5 of 150, which the statistics section says to quote as a count; its bootstrap interval, [0.7, 6.7], sits in the range where simulations of an analogous interval cover about 0.89 at a true rate of 0.03 rather than 0.95, so it may be narrower than it claims. The shipped auditor flags 16.0% [10.1, 22.3]. The stronger model blocks less of everything.

Second, the sign reverses under a weaker severity rule (Figure 3). Counting any finding rather than only findings graded BLOCKER, an exploratory rule that was not preregistered, the stronger model reaches 59.1% [47.3, 70.6] on the defect population at 34.0% [26.5, 41.7] on correct code, against the shipped auditor's 34.5% [23.6, 45.5] at 19.3% [12.8, 26.3]. The ratio of those two rates, a derived point ratio with no interval, is 1.74 and 1.79 respectively, so nothing here shows one family buying more recall per false positive than the other. Under any-finding grading the stronger model's findings cover more instances, and under the BLOCKER rule fewer, so the -26.4 depends on the counting rule. That does not show why the families' findings differ, or that any finding identifies the hidden failure.

Third, the two families are not sampled alike. The shipped cross-vendor route's capability card does not permit a temperature, so no sampling parameter is sent and the route varies between readings; the same-vendor route permits one, the layer sends temperature 0, and that route is nearly deterministic. A union over nearly identical readings cannot grow much, and it does not: over $K = 1$ to 8 the cross-vendor family gains 19.3 points and the same-vendor family gains 3.0. At one reading the gap between them is -10.1 points, not -26.4 . The rest accumulates as the two unions grow at different rates, and because model and sampling differ together, the design cannot say how much of that difference is the sampling and how much the model. We regard this as the most serious limitation of the contrast, and §D.6 records what would be needed to remove it.

A third family points the other way and belongs beside the headline. A high-reasoning model of the cross-vendor family reaches 32.7% [20.7, 45.0] union recall at four readings while flagging 10.7% [5.9, 16.1] of correct code, against the shipped auditor's 30.0% at 16.0% with eight. Paired on the same instances, the two advantages are never established together. At a common four readings it leads on recall, $+11.5$ [$+2.9, +21.3$], with an unresolved false-positive difference of -0.3 [$-3.8, +3.4$]; against the shipped auditor's full eight it leads on false positives, -5.3 [$-10.3, -0.7$] unadjusted, but not on recall, $+2.7$ [$-6.5, +13.0$] (§B.5). The evidence therefore does not support "a stronger model is a worse auditor". It supports the narrower statement that this particular same-vendor model, at this severity calibration and sampled at temperature 0, blocks far less.

A same-vendor route of the generator's own size reached lower union recall than the cross-vendor one on these instances, -12.7 points [$-25.0, -0.9$] at eight readings, signed so that a negative value favours the cross-vendor route. The interval is unadjusted, its upper end is 0.9 points from zero, and that route is sent temperature 0, so this is a direction under this sampling and not a ranking of auditors. A study on a second substrate was meant to test whether the sign reverses there; it closed without a quotable number, and on its clean run the same contrast included zero, so no reversal was established either

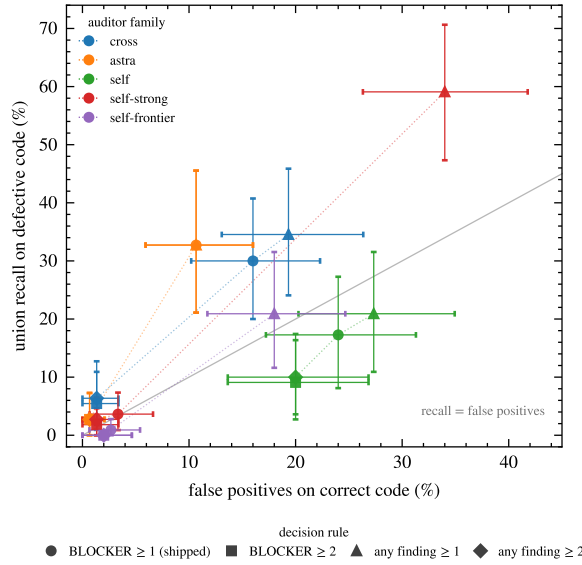


Figure 3. Union recall is a property of the decision rule, not only of the auditor. Each point is one auditor family re-graded under one of four decision rules on the same archived readings, with problem-cluster 95% intervals on both axes; no new model calls were made. The families are read at different depths: cross, self and self-strong at all eight of their readings, astra and self-frontier at their four. The figure therefore places each family at its full archived depth and is not an equal-depth comparison; the common-depth contrasts are in Table 1. The dotted guides join a family’s four points and are not a measured curve. Two things the accompanying table cannot show: the families do not lie on one frontier, and self-strong has no point under the four deterministic rules between 3.3% and 34.0% false positives, exactly where a comparison matched to the cross-vendor auditor would sit, so its ranking there is bracketed rather than measured. Nothing in this figure assesses finding content; a flag is not a diagnosis.

(§B.5). The figure above is a measurement on this substrate, and whether it is a property of same-vendor auditing is open.

B.3. One rulebook sentence moves flags, and does not measurably move outcomes

Adding a single rule to the auditor’s constitution, telling it to find what the visible tests do not cover, moved flags on the defect population from 10 of 56 to 25 of 56: +26.8 points, cluster [13.6, 40.4], exact McNemar $p = 0.0003$, cluster sign-flip $p = 0.0009$. Together with the pooled defect-and-correct flag contrast, this is one of only two comparisons that clear the Bonferroni threshold, and the two are the same effect seen twice.

The estimand is narrower than the sections above and must be read at its own scale. It is measured on 56 instances from 41 problems, not the 110-instance defect population. It is recall at a single reading, union recall at $K = 1$ on that population, because every loop arm revises once. And it is exploratory, because the preregistration named outcome contrasts rather than flag contrasts. For those reasons we do not compare its magnitude with the -26.4 or the $+19.3$ of the preceding sections, which are union recall at $K = 8$ over 110 instances.

Two costs travel with it. On correct code the same rule moved flags by +12.50 points. Every discordant pair points the same way (7 against 0), so by the rule stated in §A.3 the percentile bootstrap’s bound at zero is an artefact, and the admissible interval is the unconditional one. That interval, computed with a grid-approximated nuisance supremum and ignoring clustering, is approximately rather than guaranteed exact; it is $[-3.6, +26.8]$ and **does not exclude zero**, and its exact McNemar and cluster sign-flip p are both 0.0156, against that study’s Bonferroni threshold of 0.003125. The directed statement the records support is the pooled one, where 23 discordant pairs point one way and 1 the other (exact McNemar $p = 3.0 \times 10^{-6}$, cluster sign-flip $p = 1.0 \times 10^{-5}$): the sentence makes the auditor louder overall. On the correct stratum alone the cost is +12.50 points with an interval that contains zero.

The second cost is to outcomes. On the contrast that isolates the rule, the referent arm against the cross-vendor arm on hidden-suite pass rate after one revision (both strata, 112 instances on 96 problems), the effect is +5.36 points, cluster $[-0.89, +12.07]$, exact McNemar $p = 0.146$ and cluster sign-flip $p = 0.183$, which does not exclude zero. The rule demonstrably moves what the auditor writes down. Whether that converts into code that passes the hidden suite is not established, and since this paper argues that a flag is not a catch, that distinction applies to our own intervention first.

B.4. Most of what the auditor missed is failure the specification never determined

Fifty-seven of the 110 defective instances were flagged by no reading of the three families read at that point. That is a statement about twenty readings, not about detectability. If an instance is caught with probability p on each independent reading, twenty readings miss it with probability $(1 - p)^{20}$, which is still 98% at $p = 0.001$ and 36% at $p = 0.05$. Never flagged here does not mean never flaggable, and we use the residual as a population to characterise rather than as a set of defects shown to be invisible.

We can put a number on how flaggable it is. Reading six families at forty readings, including the same auditor under a rulebook with one sentence added, shrinks the never-flagged set from 57 to **32 of 110**, and **43 of the 68 instances both raters call specification-undetermined are flagged by some reading of some family**. Being undetermined is not sufficient for being missed. What does not move is the composition: the undetermined share of the never-flagged set is **44 of 57**, 77.2%, at three families and twenty readings, and **25 of 32**, 78.1%, at six families and forty readings. The residual loses nearly half its members and keeps its shape.

The first classification of that residual asked whether the visible tests exercised the failing input before it asked anything else, and it called 46 of 57 unexercised edges. Re-rating the same instances under a second protocol reverses that. The second protocol asks first whether the specification’s prose determines the hidden suite’s expected value, redefines the categories by entailment, and uses two raters in consensus; ordering, definitions and raters all changed together, and the study cannot isolate which mattered. The two raters agree that 44 of 57 (77.2%, cluster [62.1, 91.1]) are cases where the prose does not settle the value, and that 2 of 57 are cases where it does and the visible tests merely never built the input. The preregistered condition for withdrawing “the residual is mostly unexercised edge” fires on both of its clauses, and that claim is withdrawn. The interval is conditional on a category and has no committed coverage simulation, so it is uncalibrated, as is every category-conditioned interval we report.

A classification that reverses when its protocol changes is, by itself, weak evidence about the auditor, and the raters are not independent of it: one was the agent that ran this programme (§A.4), which wrote the rubric and knew the earlier labels and which claim was at stake. Three things support the second reading. The rubric was fixed before any label was written. The labels were committed before we looked for outside evidence. And when a first pass was found to have carried instance identifiers and a sheet name into the raters’ view, the labelling was re-run with identities stripped and the result held. Where an outside party had independently recorded that one of these specifications is defective, in twelve MBPP tasks named in a paper written for another purpose, our raters agree on 9 of the 11 that fall inside our defect population, with both disagreements on the boundary that paper’s own “incomplete” category draws. That check is one-sided by construction: it can support agreement where an outsider looked, and can bound nothing about the rate.

A rater outside the residual’s families, asked a different question. One of the two raters above is *gpt-6-astra*, whose own misses help define the residual. A model that is not one of the families defining it, *gpt-5.6-luna* (which read these instances only as a secondary audit route in an earlier architecture study), later rated the same 57 instances post hoc under a different rubric: whether the prose determines an answer at all, on a sheet showing at most three failing inputs and never the expected values (§B.5). It labels **38 of 57 undetermined**, 13 determined and 6 cannot-tell. The two figures measure different constructs. The earlier category also admits prose that determines an answer the oracle contradicts, which this rubric does not count as undetermined, so 38 is not a corrected estimate of 44. What they share is that a participating and a non-participating rater both put a majority of the residual on the undetermined side, and the second agreed with itself on only 7 of 11 repeated items.

B.5. Eight studies, and what they license after between three and eleven review rounds each

Eight further studies cleared independent review after between three and eleven rounds each. We report them here rather than folding them into the sections above because several are narrow or negative results whose limits are part of the claim, and because the shape of their refusals is itself a finding about this method.

A rulebook sentence moves a second vendor’s family. The referent rule, *find what the visible tests do not exercise*, raises union BLOCKER recall on *self-strong* (Sonnet 4.6) at four readings by **+20.9 points**, cluster [+10.8, +31.8], at a false-positive cost on correct code of **+8.0 points** [+3.4, +12.7]. The grading rule shows no demonstrated improvement, +0.9 points [−1.8, +4.5], and that hypothesis fails its registered criterion, which demonstrates neither equivalence nor that the rule did nothing. This reproduces §B.3’s effect on a different vendor’s model at a different K , with its own price. A post-hoc quantity from the same study, whether a flagged finding *reports a failure* on the named class, is **3 of 110** under the

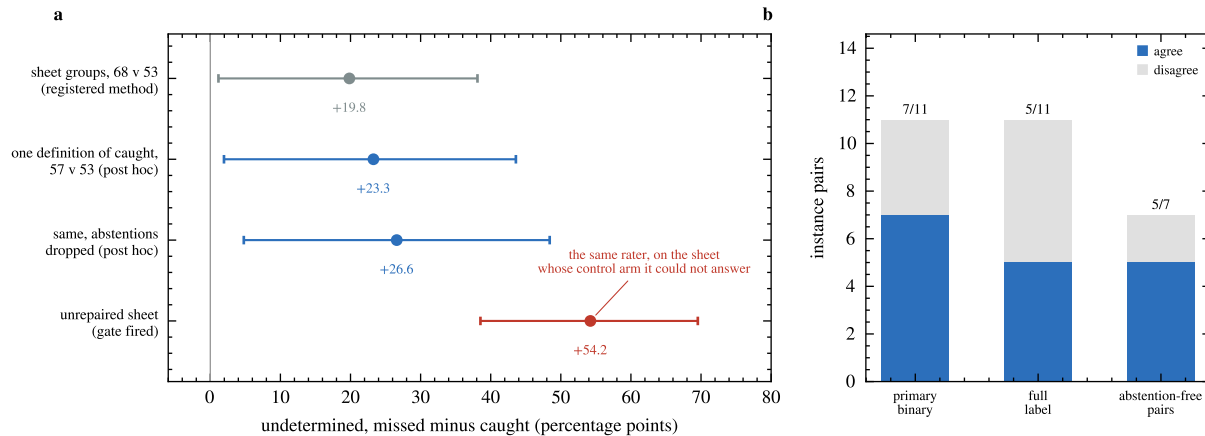


Figure 4. The residual contrast, printed beside the instrument that produced it. a, The share of instances a rater judged specification-undetermined, missed group minus caught group, under each reading, with problem-cluster 95% intervals. The two sheet groups use different definitions of *caught* and share eleven instances, so the two readings that apply one definition are the ones that compare populations; both are post hoc. The red point is the same rater on the same sheet before its control arm was rebuilt, when 42 of 53 caught entries carried no failing-input evidence and all 42 drew *cannot-tell*. With that arm the contrast was larger and pointed the way the hypothesis points. The repair and a fresh model run changed together, so presentation, sampling and batch context are not separated, and a preregistered abstention gate excluded the earlier figure. **b,** The same rater's agreement with itself on the eleven instances the sheet carries twice, with identical text and identical evidence. The contrast in **a** is read off the instrument in **b**, and the claim this paper draws from it is stated under that constraint.

shipped constitution and **27 of 110** under the rule, and it may not be quoted as defect recognition. Its rubric admits a finding even where it grades the class non-blocking or says the specification is silent, so findings that disclaim the requirement are counted in.

A measured 60.9% is not a higher ceiling, and it is the same estimator as the 30.0%. On the defect population the same rule takes union recall (the identical quantity: blocked by at least one BLOCKER, unioned over eight readings) from 33 of 110 to **67 of 110 = 60.9%** [48.6, 72.5], **+30.9 points** [19.1, 43.1], at **+20.7 points** [12.8, 28.8] of false positives, with a single-draw rate of 23.6% [17.1, 30.3] that fails the product bar on all eight draws. The caution that a flag is not a catch attaches to both figures equally. What separates them is their price, 36.7% [28.5, 44.8] false positives against 16.0%, which is why the shipped route is the shipped route. The fitted asymptotes differ by +27.1 points [10.9, 39.0], and *that does not establish a higher ceiling*. The cross-vendor arm never flattened, so one of the two terms is an extrapolation. Heterogeneous low-probability detection imitates a ceiling at this budget, since an instance found with probability 0.02 per reading survives eight readings about 85% of the time. And this study's own zero-inflated estimator finds no never-detectable class in *either* arm. Its adjudication kill is arithmetic rather than a labelling artefact: only 9 of 110 instances drew any finding, so the largest defect-asserting count available under perfect adjudication was 9, below the floor of 20. Every adjudication-derived figure in it is conditional on labels whose production can no longer be audited, because one rater's raw reply, prompt, launcher and execution log lived only in a session scratchpad and were destroyed between the review that flagged them and the repair.

Corroboration does not validate a generated test. A wrong generated test, one that fails on the dataset's canonical solution, is not reliably recognised by asking whether further independent generation draws also fail on the same candidate. Neither paid rule reaches the preregistered threshold on the registered population, which covers failing applications on all three candidate strata, 58 of the 107 of them on candidates that already fail their visible tests. The retained wrong-test rates are **12.2%** (Wilson 7.0–20.6; cluster 3.6–22.6) and **14.9%** against a limit of **2%**, and neither is shown to improve on the free within-draw comparator at 12.8%: the better paid rule keeps the same wrong applications plus four correct ones, a small observed advantage the study cannot resolve. On the candidates a product would meet, those passing their visible tests (post hoc), the rates are 6 of 33, 18.2% [0.0, 44.8], and 10 of 43, 23.3% [6.8, 47.1]. The first leaves its relation to the 2% threshold unresolved; the second's unadjusted interval lies wholly above it, which is exploratory evidence of exceeding it. Neither is a preregistered test on that population, whose registered kill rule was a point estimate on the mixed population. A floor of 2.4% holds on the original unique-test quantity because 29 of the 45 wrong tests never fail on any candidate at all. The claim is negative and bound to this substrate: these rules do not validate generated tests here, which is not to say generated tests cannot be validated.

An injected population does not give the residual result a prospective test. We constructed a population of injected defects intended to make specification-determinedness hold by construction. It does not. The filter registered to recover each edit’s first failing hidden input is a truthiness check on a field the injector wrote: of 281 recorded witnesses, 179 are strings and 102 are lists, and none is executed. Four of the remaining five filters accept cases their registered wording excludes, such as a suite that reports success after an early exit, a process that terminates without an assertion or an exception, and an added import from a module already imported. No filter establishes specification entailment, and repairing the first one alone would not have: recovering the failing input and executing a witness shows that the edit changes behaviour on an input the hidden suite exercises, not that the specification entails the value expected there. What may be quoted is descriptive. At eight readings the auditor blocked **90 of 92** accepted injected instances against **11 of their 92** unmodified twins, 79 discordant pairs one way and none the other, and against **33 of 110** on the natural residual. The instances the construction’s own gate *refused* were blocked **31 of 40**, which is why that gate cannot be assumed conservative. A probe separates the injected set from the natural one at 96.7% on the items it answered, *by a feature it does not identify*: the two arms share no problem, so task composition alone could produce the result, and the probe’s model is one of the two gate models. §B.4 therefore keeps exactly the status its own study gave it, post hoc and unreplicated. A successor needs an independent party to validate the actual violation *and* the specification-entailing expected behaviour under a fixed, blinded protocol. Neither a second substrate nor an independent rater suffices alone, because task composition and construction confounding survive both.

A study on a second substrate closed without a quotable number. In a second benchmark, the visible-test text shown to the auditor *and* to the generator was sliced out of its test class without the class header, so it failed to parse on all 300 tasks while scoring executed the intact class. The repair for that fault introduced a second one, re-attaching a hidden method’s decorators to the first visible method on 26 tasks. A third run on a clean extraction reproduced five of the six findings that the second run had reversed and did not reproduce the sixth. The criterion registered before those numbers existed was stated over the set, so the strict reading applies and the study is closed after three runs, about \$82, and no quotable number. Whether five of six should have licensed the five is recorded as an open question in CLAIMS.mcd.

Measured union recall is rule-dependent, and a headline’s magnitude belongs to its own rule. Re-grading findings already archived by two of the studies above under four decision rules, with no new model calls, leaves the shipped BLOCKER threshold’s estimate intact for its own rule, which *was* preregistered as “ ≥ 1 BLOCKER finding”, while *reversing its sign* under any-finding grading. That reversal is a mechanical consequence of changing what counts, and it identifies nothing about why the families produced different findings or different severities. At a common depth of four readings, *astra* exceeds the cross-vendor auditor on union recall over the defect stratum by **+11.5 points** [+2.9, +21.3], paired within instance, with a false-positive difference of -0.3 points $[-3.8, +3.4]$. The two sides are not averaged alike. The cross-vendor arm has eight readings and is averaged over all $\binom{8}{4} = 70$ four-draw subsets, while *astra* has exactly four, so $\binom{4}{4} = 1$ and its estimate is a single realisation with none of its reading-selection variance averaged out; one arm’s luck in four draws is fully in the number and the other’s is not. On the cross-vendor auditor’s complete eight-draw ladder the same contrast is **+2.7 points** $[-6.5, +13.0]$ with a false-positive difference of -5.3 points $[-10.3, -0.7]$. That comparison points toward lower false positives, not higher recall, on an interval whose upper end is 0.7 points from zero and which is not adjusted for the other contrasts computed beside it, so it is a direction the data lean and not one they establish. Observed false-positive rates near one another are not a matched operating point, and one family has no point under the four deterministic rules between 3.3% and 34.0% false positives, precisely where a matched comparison would sit, so its ranking is bracketed rather than measured. Nothing here assesses finding content against the hidden failure, so none of it speaks to what any finding identified.

The ambiguity gap survives a rater that did not write the rubric, and the instrument that shows it disagrees with itself.

The contrast in §B.4 between missed and caught instances had been rated by two models, one of them the agent that wrote the rubric. A third model re-rated the whole sheet after we rebuilt it on mechanically recovered witnesses. That rebuild was necessary because the control arm had been unanswerable: 42 of 53 caught entries carried no failing-input evidence, and all 42 drew *cannot-tell*. The share labelled *undetermined* is 43 of 68 against 23 of 53, **+19.8 points** [+1.2, +38.1]. Those two groups are not two populations, because they define *caught* differently and 11 instances fall in both; removing those 11 from the missed group gives **+23.3 points** [+2.0, +43.6], post hoc and larger, so the overlap had understated the contrast. The limit is part of the result. On those 11 instances, shown with identical text and identical evidence, **the rater agreed with itself on 7 of 11** on this binary outcome and 5 of 11 on the full label, where the earlier study’s two raters each agreed 11 of 11 (Figure 4). Every interval’s lower end lies within five points of zero, and this remains a model; no human has rated this material. The *unrepaired* sheet scored **+54.2 points** [+38.5, +69.5], 2.7 times the repaired contrast and in the direction that flatters the hypothesis. That is a descriptive association between one repair and one fresh re-rating, not evidence about

specifications: the repair, the run and its sampling changed together, and a preregistered inconclusiveness gate catches it.

Adding the missing rule to the specification moves diagnosis, but only where the rule matched the oracle. The contrast above conditions on the specification; this study changes it. The population is the 44 never-flagged instances that both earlier raters called specification-undetermined. For the 32 on 19 problems that passed the generation gates, a model shown the failing *inputs*, but never their expected values, wrote one behavioural rule into the prose, and an auditor re-read the same code and the same visible tests under the original, the clarified and a same-length placebo specification, four readings each. An instance counts as diagnosed only if both adjudicators judged some finding to state the behaviour the hidden suite expects. Diagnosis rose from **0 of 32 to 9 of 32** under clarification and stayed at 0 of 32 under placebo, **+28.1 points**, problem-cluster 95% interval [+7.7, +51.4], which meets the registered criterion. The registered secondary, on the 23 instances where study 21’s consensus and the third rater’s label agree, is 8 of 23 against 0 of 23, +34.8 [+10.0, +60.9].

Given the observed data, that criterion added little. Every discordant pair points one way, so no bootstrap resample can fall below zero, and with successes on five of 19 problems a resample is exactly zero only when it misses all five, with probability $(14/19)^{19} \approx 0.003$; the registered run has 27 such resamples in 10,000. Once five success-bearing problems are observed, the interval’s exclusion of zero adds little to them (with one such problem it would not have excluded zero), and §A.3 says to read a one-signed contrast from an exact test instead. The registration kept the bootstrap and we report it as registered, but the informative figure is the cluster sign-flip test the registration prefers. The nine successes sit on five problems, and it gives $p = 0.060$ (exactly $2/2^5 = 0.0625$ on enumeration), against 0.0039 for a McNemar test that counts nine independent pieces of evidence where five exist. It is not significant at 0.05. The registered naming-rate secondary was never computed.

Two further limits are part of the result. *The inferred rule contradicts the hidden suite entirely on 23 of the 32 instances and partly on 2*, and diagnosis was 0 of 23 where it contradicted against 9 of 9 elsewhere. That split was classified after the outcome was seen, not assigned, and the agreeing instances may simply be the easier problems; it does not say which property of the edits moved diagnosis. And a separate model rated the clarified prose more determining than the original by +28.1 points [+2.6, +54.3] but than the placebo by +18.8 [−9.4, +45.2], so whether the placebo was inert as a specification is unresolved. One adjudicator is the agent that ran the study and the other is *gpt-6-astra*, the model that also reviewed this study (§A.4); they agreed on 84 of 91 items, $\kappa = 0.85$, which is agreement and not accuracy. The study supports a specification-edit contrast on these selected instances. It does not establish that underdetermination caused the original misses, and §B.4 is not strengthened into that claim by it.

What a flag on correct code points at. The false-positive rates above count flags on code that passes its hidden suite. A suite may leave behaviour unexercised, so a flag there need not be a false alarm. A preregistered study re-read the 150 correct instances with the shipped route at eight readings, keeping the finding texts ceiling 1 had discarded; 20 were flagged (13.3% [7.9, 19.2], a fresh sample of the route’s 16.0%). A model given each finding, but not the reference solution or the hidden tests, extracted the concrete inputs it names, and candidate and reference were run on them. **Two of the 20 disagreed with the reference** on an extracted input the hidden suite does not exercise (10.0%, problem-cluster [0.0, 26.3]); on nine they agreed, and nine yielded no valid input. Because the interval includes zero, the registered reading is that at this resolution there is no evidence that this route’s flags on test-passing code identify reference disagreements. No class licenses the words “true positive” or “false alarm”: a disagreement does not show that the specification was violated, and agreement on an extracted input does not show a finding wrong.

What the refusals were about. These eight studies cleared review at rounds 3, 3, 4, 6, 7, 9, 10 and 11, fifty-three rounds in all. Most of those rounds ended in a refusal of the *repair* rather than of the measurement. Measurements moved on review in three of the eight: the sweep’s first review found its loader reading one family’s draws 4 to 8 as 1 to 5 and another’s population from a partial draw, and every cross-family figure was recomputed; a label count in the third rating’s report was 71 where its data say 67; and the clarification study’s primary moved by one instance and three classification rows (§A.4). Every figure above reproduced independently at every round after its last repair. The recurring failure was a check or a sentence that claimed more than it delivered. A binding protected one sentence of a withdrawal spread across six; when it was rewritten to protect all six, two of them were truncated before the words that carried their meaning. One check stripped HTML comments and called the result what a reader sees, and one test was named for a code path that never called it. A second pattern appeared in the last thirteen rounds: most of those refusals were *overstatements away from our own interest*, such as a registration whose chronology its own repository contradicted, a withdrawal that overcorrected past what the record supported, a publication prohibition a registered gate never imposed, and a limitation credited to another study’s work that the other study had not performed. Modesty is not the safe direction; it is another direction in which a sentence can be

wrong. What corrects it is a smaller claim written where the claim was, and a heading, a title and a limitation are all claims.

B.6. What follows for a recall number

Taken together, the results above say an audit recall figure is a joint property of the auditor and its material. It moves with which route reads (model and sampling together, which this design does not separate), with what the rulebook tells it to look for, and with the severity threshold at which a finding becomes a block, which is rarely reported. What it misses is associated with whether the hidden suite asks a question the specification answers. That association is observed in the residual and in a specification-edit contrast on selected instances; it is not isolated as a cause.

C. The science studies in full

Acts 2 and 3 measured the model tier on short general-purpose functions. A scientist would hand an auditor three different jobs: check the code that computes a result, check that a reported result follows from that code, and check the data an analysis starts from. We ran one preregistered study per job, each with ground truth that no model supplies. Scientific code (§4.1) is judged by scientists’ numeric tests, results (§4.2) by re-execution, and data (§4.3) by the faults we injected. Every item in all three is audited through the product’s own audit call with its shipped rulebook, and the same flag rule holds throughout: an item is flagged when the model grades any finding BLOCKER.

C.1. Scientific code

Substrate. SciCode (Tian et al., 2024) decomposes 80 research problems from 16 subfields into 338 steps, each a function with numeric tests written by scientists. Its targets are released separately from the code, which lets the tests act as a hidden oracle. On the benchmark’s development split, the gold code passes 48 of 50 steps in our environment; the two steps whose gold fails here are excluded, as are the three the official harness skips.

Population. The generator of Acts 2 and 3 wrote every step in SciCode’s own protocol, three samples per step, each sample building on its own earlier steps. A candidate is *correct* if it passes the step’s tests and *defective* if it fails or times out. A step enters either stratum only if at least one sample passes it, so a failure counts only where this environment has shown the step can be passed. This yields 81 defective instances from 30 problems and 374 correct ones, from which 150 (53 problems) were drawn by seed.

What differs from Act 2. SciCode has no visible and hidden split, and we did not invent one. The auditor sees the problem, the step, its background, the function header, the allowed dependencies and the candidate, and no tests of any kind. The shipped cross-vendor auditor and the generator’s own model each read every instance eight times. The generator’s model is reached through a command-line route rather than the API; the route adds a short fixed context and uses default sampling, so this study’s same-vendor family is not temperature-0, unlike Act 2’s.

Two arms. After the registered arm’s outcomes, a second registered arm changed only the task text, naming the step’s function as the deliverable; the program bytes are identical.

Residual. Defective instances flagged by no reading of either family are rated on a blind sheet under Act 3’s question adapted to science: do the step’s prose and background determine the value the test expects, including tolerance, units, conventions and numerical method. The raters are the agent that ran the study and a model that audits nothing here.

Results. In these runs, what the auditor was told the deliverable is changed its flag rate substantially (Figure 2a). In the registered design the task opened with the whole multi-step problem, and the shipped auditor flagged 87.7% [72.7, 98.6] of defective instances and 68.7% [56.6, 79.7] of correct ones at eight readings. Many of its objections to correct code were that one step did not implement the whole problem. A second registered arm, run after the first and changing only the task text, named the step’s function as the deliverable and gave the problem description as context. On the same program bytes, flags on correct code fell to 36.0% [26.4, 45.6], a paired change of -32.7 points $[-42.8, -22.4]$, and recall fell to 72.8% [59.3, 87.3], -14.8 points $[-28.6, -2.7]$. The generator’s own model flagged less of both strata in both arms, 46.9 points less of the defective instances than the shipped auditor (exact McNemar $p = 1.6 \times 10^{-9}$ in each arm). With no tests shown, the shipped auditor discriminates, but modestly: at one reading, 61.7% of defective against 19.8% of correct instances.

On reading a sample of its flags on correct code, some point at real divergences between the code and the step’s text or its own documentation, and some at contradictions inside the benchmark’s own instructions or at tests too weak to tell outputs apart. Few defects went unflagged by any of the sixteen readings: 7 in the first arm and 19 in the second. Most of them the

two raters called undetermined by the prose, 4 of 7 and 13 of 19 (for example an entropy’s logarithm base, or the direction of a minimum-image vector), but the registered test that this share exceeds the general-code residual’s 77.2% is killed in both arms by its own interval rule. These labels describe the prose, not why a defect was missed.

C.2. Scientific results

Items. Each correct SciCode instance whose output on one seeded test case is a real scalar or a real array of at most ten elements becomes a small results increment: the program, the test’s inputs, a script that runs one on the other and writes a run log, a `results.json` citing each value’s log line, metadata declaring every input at a revision, and a one-paragraph report whose numbers carry the product’s number→source annotations. Seventy-four of the 150 correct instances qualify, from 36 problems. Each yields one clean item and one faulty item. The faulty item carries one of seven faults, assigned in rotation by seed. Five are *reporting* faults, where the artefacts disagree with each other or with the log the code wrote: a $\times 10^3$ scale slip, a sign flip, a value from another test case, a 5% relative error, and a transposed digit pair in the report alone. Two are *fabrications*, where the scale slip or the 5% error is written consistently into the log, the results and the report, so that every artefact agrees with every other and none agrees with the code.

Three auditors. The LLM auditor reads the increment four times, with the project’s checks set to the shipped `science` profile. The model’s prompt includes the deterministic result, as it does in the product, so on reporting faults the model has seen the deterministic finding. The deterministic profile’s own verdict comes from the same call. Re-execution runs the program on the inputs and compares the output with the reported values.

Gate. Before any model call, every clean item had to pass the science profile with no BLOCKER and re-execute to its reported value; all 74 did when first built. A two-reading pilot then found an instrument defect: no file in the item performed the run the report described, and the auditor blocked on that, correctly. We added the run script, discarded the pilot readings, and added a third condition to the gate, that each clean item’s run script reproduce its run log byte for byte. All 74 passed all three conditions with no repair.

Results. The deterministic profile does exactly what it was built for (Figure 2b). It flags all 49 reporting faults, because the log, the results and the report no longer agree and number→source says where, and none of the 25 fabrications, because in those everything agrees with everything else. It flags no clean item. The LLM auditor flags 23 of the 25 fabrications at four readings, 21 of them with a sound finding about the fault (post hoc). Those findings work the result out from the code and inputs, for example evaluating a KL divergence of two small distributions to 1 where the report says 1000. The other two rest on calculations that are themselves wrong. Paired on the faulty items, all 23 discordant items go the LLM’s way (exact McNemar $p = 2.4 \times 10^{-7}$; a problem-level sign-flip test, post hoc, gives 3.8×10^{-6}). On the reporting faults its rate is of the combined workflow, since its prompt carried the profile’s findings. Re-execution flags every fault except four whose outputs are within `np.allclose`’s default absolute tolerance of zero; two of those outputs are floating-point zeros.

The LLM also flags 12 of the 74 clean items, and 8 of those flags are right. In 9 instances the benchmark passes the test’s inputs inside the call itself, so the report’s sentence that the program ran “on the inputs in `inputs.py`” was false; 7 of those clean items were flagged for it. Two of four outputs whose unit the solution documents were flagged for our `dimensionless` label. The remaining 4 flags rest on wrong calculations or, in one case, on a syntax error that is not there. On this evidence a provenance chain and a reader that recomputes are complements: the first is exact about agreement and blind to consistent error, and the second sees consistent error but is sometimes wrong itself.

C.3. Scientific data

Substrate. Four physical-science tables from the UCI repository, each under CC BY 4.0: concrete compressive strength (Yeh, 1998), airfoil self-noise (Brooks et al., 1989), combined-cycle power-plant output (Tüfekci, 2014), and superconductor critical temperature (Hamidieh, 2018). Exact duplicate rows and physically impossible rows (relative humidity above 100%) were removed from each pool before any item was drawn.

Items. Each item is 120 rows sampled from one pool with a data card that gives every column’s meaning and unit and, where the source states one, its range. There are 35 clean items per dataset and five per dataset for each of seven faults: mixed units in 20% of one column’s rows, three negated values of a non-negative quantity, 12 duplicated rows, the target shuffled among 30% of rows, two columns with disjoint ranges swapped, a `−999` sentinel in 5% of one column’s cells, and a continuous column rounded to integers.

Auditors. Both model families read each item four times, with the dataset as the deliverable and the card as its documentation.

The comparison is a deterministic validator written from the card alone and frozen before any item was built: declared ranges, non-negativity, duplicates, sentinels and integer-valued continuous columns. By construction it catches the negations, duplicates, sentinels and rounding. It sees mixed units or swapped columns only when they push a value below zero or out of a declared range, and a shuffled target not at all. The question is whether the model catches what the validator cannot. Two instrument defects were found and fixed by amendment before the readings reported here: the card and file named one column differently, and the files first sat outside the audited scope, so the product had told the model there was nothing to audit.

Results. The two tiers catch different faults (Figure 2c). The validator flags 92 of 140 faulty items (65.7% [57.9, 73.6]) and no clean item. The shipped auditor flags 65 of 140 (46.4% [38.6, 54.3]) at 4 of 140 clean items (2.9% [0.7, 5.7]), and their union flags 112 of 140 (80.0% [73.6, 86.4]) at the same 4. (These intervals resample items within dataset; four datasets are too few to resample, so no interval here carries between-dataset variation.) Of the 20 faulty items the auditor flags and the validator misses, 18 carry a finding about the injected fault itself, 9 of them mixed units and 9 swapped columns (post hoc). In one, the auditor reasoned that three atoms spanning a 135 pm radius range cannot have a mean radius of 1.6 pm, which is the cell the fault had converted to ångströms. The other 2, and all 4 flags on clean items, are requests for fuller column descriptions on one dataset’s card. The auditor flagged no shuffled target and no duplicated rows (0 of 20 each), and on every fault the validator was written to catch it did worse than the validator. The generator’s own model, read through its command-line route, is not a usable data auditor here: it flags 67 of 140 clean items, and in at least 48 of those one finding states a row count, field count, missing column or truncation that the file contradicts (post hoc lower bound).

The deterministic validator is not neutral. It was frozen from the card before any item existed, but it omits one range the concrete source’s documentation states and uses a list of continuous columns the card does not give; with the omitted range it would flag 93 of 140. Unit conversion also left floating-point tails that mark some converted cells in two of the tables. None of the auditor’s mixed-unit findings cites them, but that does not show they had no influence.

D. Discussion in full

D.1. What a union-recall number means

A union-recall figure depends on the false-positive rate it cost, so recall quoted without that rate cannot be carried to another setting. We make this as a methodological point. The measured support we tried to give it came from a second substrate that was run three times and closed without a quotable number (§B.5). The point needs no second substrate: recall and false positives move together with any severity threshold, and a critic can be tuned to any recall by blocking more. Our records support a narrower version on the substrate we report. The same auditor at two severity rules sits at two different operating points (§B.2), and the shipped auditor’s 30.0% [20.0, 40.7] recall is quoted only beside the 16.0% [10.1, 22.3] it cost.

The cleaner experiment for the general claim is a severity sweep within one substrate. Our sweep re-grades archived findings at four decision rules and shows each family’s recall and false-positive rate moving with the rule (Figure 3). Its rules are deterministic and do not place the families at a common operating point. A randomised mixture of two archived rules could reach one in expectation without new model calls, but it would be a new post-hoc policy needing its own uncertainty analysis, and we have not built it. Two substrates differ in specification length, solution length, problem population, defect distribution and suite construction all at once, so comparing them could never have isolated the operating point.

D.2. Repeated reading buys nothing where the verdict does not vary

Union-of- K is the instrument this paper uses to look for a ceiling, and part of what it cannot measure follows from arithmetic alone. A union over readings that do not differ cannot grow, so the curve is flat for any route that returns the same verdict however often it is asked.

Our records establish the design fact that makes this matter here. The same-vendor routes are sent temperature 0, while the cross-vendor route is sent no sampling parameter (§B.2). From $K = 1$ to 8 the first family’s union gains 3.0 points and the second’s gains 19.3. No temperature-matched arm was run, so we cannot say how much of that gap is sampling and how much is the operating point. We treat it as a confound on our own contrasts.

For best-of- K , union-of- K or panel aggregation over critics, what follows is the arithmetic: a route whose verdict does not vary between readings gains nothing by being read again. We measured per-instance flag counts, not any distance to a decision boundary, and ran no panel aggregation, so we offer no law for how much a varying route gains. The diminishing

shape of the curves is not itself evidence of a ceiling either. Exact subset averaging makes each reading’s marginal gain no larger than the last by construction, and the curves measure the size of those gains.

D.3. What the ceiling is made of

We do not rank these contributions, because they were measured on different populations, depths and rules. Two of them share a scale. Eight readings of one auditor buy 19.3 points of union recall. Twenty readings across three families buy a pooled 48.2% [36.7, 60.0] at 36.0% false positives on correct code, and leave the complement, 51.8% [40.0, 63.3] of the defect population, untouched. Changing which model reads moves the same quantity (union recall at $K = 8$ over the same 110 instances) by 26.4 points, more than repetition’s 19.3. A same-size same-vendor substitution moved it by only 12.7, and §B.2 gives three reasons not to read either number as auditing ability. Changing one sentence of the rulebook moves flags by 26.8 points, on 56 instances at one reading, which is not the scale of the other two.

A fourth contribution is not the auditor’s at all, and we can describe it only post hoc. Under a rule that asks whether the specification’s prose determines the hidden suite’s expected value, 44 of the 57 never-flagged defects fail. We tried to build a population where the prose does determine the answer. Review established that no filter in it enforces that property, so it licenses description only (§B.5).

The association between never being flagged and never being determined is empirical. Selecting the missed instances does not fix their labels, and against caught instances the third rater’s post-hoc contrast is +23.3 points [+2.0, +43.6]. That rater agreed with itself on only 7 of 11 repeated items, and the interval is conditional on its recorded labels. Selection and confounding rule out a causal reading, so the contrast locates the misses and does not estimate how much the benchmark contributes.

One intervention changes the specification instead of conditioning on it (§B.5). Adding a behavioural rule to the prose moved diagnosis on 9 of 32 such instances, all of them where the added rule agreed with the hidden suite and none of the 23 where it contradicted it. This is the only evidence here that manipulates the specification. Its population was selected for having been missed and called undetermined, and it cannot separate a specification that settles the question from one that settles it correctly.

The constructed population has two further limitations, one measured and one found in review. A frontier model separates its edits from natural ones at 96.7% on the items it answered, and between 79.9% and 97.3% once the items it would not classify are counted at either extreme. It does so *by a feature it does not identify*: the two arms share no problem, and the model is one of the two that gated the population. And because the construction does not enforce specification-determinedness, it licenses *no* bound on the measured ceiling.

D.4. Flagging is not naming

An audit that blocks an increment for the wrong reason has still blocked it, and a recall number counts that as a catch. For one reading of the shipped auditor the finding texts were kept and adjudicated blind by two raters. The auditor returned a finding on 9 of 110 defective instances and asserted the actual failing behaviour on 5 to 7 of them, depending on how the two raters are combined. The raters agreed at $\kappa = 0.391$. We report that beside the rate because a moderate agreement is part of what the rate is worth. That study’s preregistered kill condition fired on this result, and §B.5 states its headline in the form the kill requires. The figure describes one reading. It is not a naming rate for the eight-reading union, which would need the texts of all eight readings, and those were not archived.

D.5. What the science studies add

The science studies give the design rule of CrossAudit’s science profile (the model names evidence, code verifies it) an empirical reading, and a boundary. On results (§4.2), number→source verified every inconsistency between a report, its results file and its log, and it could not see a result that was wrong consistently. The model, by recomputing, flagged 23 of 25 such fabrications, 21 with a sound finding. On data (§4.3), a validator written from the card caught what a rule can state (negative masses, sentinels, duplicates, lost precision) and missed what needs reading across columns and units, which the model caught in 18 of the 28 such items the validator missed, at a clean-item cost of four remarks about a sparse card. In both, the deterministic tier flagged none of the clean items and the model tier carried the errors. On data the union of the two was better than either part; on results the model’s flags already contained the profile’s, whose findings its prompt carried.

What neither study shows is that the model tier scales. Both used small artefacts: outputs of at most ten numbers, tables of 120 rows. The model’s fabrication findings are hand calculations, several of them wrong, and it missed every fault that needs comparison across many rows (shuffled targets, duplicated rows). A model reading is a useful second tier where the artefact fits in the reader’s working memory, and it is not a substitute for executing the code or computing over the data.

A second observation cost us more than it gave. In all three studies the auditor flagged defects in our own construction: a card and file that named a column differently, a file placed outside the audited scope, a report that claimed a run no file performed, a provenance sentence that was false for inputs passed as literals, a unit label that contradicted the code’s own documentation, and a task that named the whole problem when only one step was delivered. Each was a correct flag on an item we had built as clean. We caught two in pilots, one when every verdict came back as an escalation, and the rest only by reading every flag on a clean item, which the registrations required. An evaluation of an auditor has to establish that its clean items are clean by the auditor’s standard, not only by the benchmark’s, or its false-positive rate measures the evaluator.

D.6. Limits

1. **Sampling differs between families.** One route is sent temperature 0 and the other no sampling parameter, which bounds how far one union can grow before any difference in ability is measured. Every cross-family union-of-K contrast here inherits this, and it is the confound we would fix first.
2. **No matched operating point.** No auditor is compared with another at a matched false-positive rate. Our sweep re-grades archived findings at four deterministic decision rules, under which one family has no point between 3.3% and 34.0% false positives. We built no randomised policy between them, so the paper cannot say whether the families differ at the same operating point.
3. **Interval coverage.** The primary intervals are problem-cluster percentile bootstraps over 56 clusters. A simulation of an analogous single-rate interval (600 resamples rather than 10,000, tested rates only) covers near nominally at the headlines’ rates and anticonservatively below about 0.10 (§A.3). For the paired contrasts and conditional shares coverage is unmeasured, and no sample size was justified in advance by a detectable-effect calculation.
4. **Substrate.** The substrate is short, self-contained Python functions with executable hidden tests, so nothing here measures auditing a change to a large program with no oracle. Both benchmarks are old enough that memorisation cannot be ruled out.
5. **One generator.** The defect population is one model’s failure mode.
6. **Raters.** The residual rubric’s raters were two models, one of them the agent that wrote the rubric, which was not blind to the earlier labels. One external label set agrees with them across eleven overlapping tasks.
7. **The constructed population.** It did not enforce specification-determinedness, and its edits are separable from natural code at 96.7% of the items the probe answered (79.9–97.3% once unanswered items are counted at either extreme) by something the probe does not identify.

Each limit is recorded in the study that carries it, together with the number it would change.

D.7. What we would do next

The sweep re-grades archived findings under four rules and shows that the auditor contrast depends on the rule (§B.2). Re-grading only moves a cut through findings already produced, so it cannot bring every family to a matched operating point: one family has no point under the four deterministic rules between 3.3% and 34.0% false positives. The experiment this paper most needs is to re-ask, instructing each family at graded severity and reading afresh, which can place the families at a common false-positive rate and decide whether the auditor contrast survives there. Matching the sampling parameters across families would remove the confound above, and reading the cross-vendor family past eight times would turn a statement about our budget into one about a curve.

We ran a first version of the experiment we most wanted, holding the code and the visible tests fixed and varying only the specification (§B.5). Two gaps remain, and they decide what it means. The clarifier inferred the missing rule from failing inputs and was wrong on most instances, so the rule’s correctness was measured, not manipulated. Writing the rule from the oracle for some instances, and a deliberately wrong rule of matched length for others, would separate a specification that settles the question from one that settles it correctly. Its manipulation check also did not show the placebo inert, and a

placebo rated no more determining than the original would. Both are cheap, because the code, the tests and the adjudication rubric already exist.

Beyond those: adjudicating more than one reading of the shipped auditor would let a naming rate be compared with a union flag rate at the same K . A second generator would show whether the residual’s character belongs to the benchmark or to one model’s failures. A population of natural defects independently known to be specification-determined would separate the injected population’s artificiality from the property it was built to isolate. And a substrate whose ground truth is not a hidden test suite would test the assumption every ceiling here rests on, that a test suite asked the right question.

E. Limitations in full

Substrate. Acts 2 and 3 measure function-level Python tasks from two code benchmarks, and Act 4 one scientific-code benchmark, small results artefacts built from it, and four tabular datasets with synthetic faults. Nothing here establishes that the same ceiling, or the same explanations, hold for another language, for repository-scale changes, for code whose correctness is not decided by an executable suite, or for scientific artefacts larger than a model can read whole. The science studies’ faults are synthetic, one generator wrote every program, and each study used one auditor model at its default sampling.

The ground truth is a test suite, and a test suite is not a specification. Hidden suites encode a reference implementation’s behaviour, including behaviour its prose never fixed. In the intervention study reported in §B.5, a generator shown an instance’s failing inputs inferred a behavioural rule that *contradicted* the hidden oracle on 23 of 32 instances. The study cannot say whether that reflects the specifications or the generator’s inference from inputs alone; either way it bounds what any recall figure measured this way can mean.

Interval coverage, measured only in part. The primary intervals are percentile bootstraps over problem clusters. For a *single rate* only, simulated under the design’s own cluster profiles and an assumed within-problem correlation, with 600 resamples rather than the 10,000 the analyses use (400 replicates; Monte Carlo standard error 0.01–0.02), a nominal 95% interval covers between 0.93 and 0.97 at the tested true rates from 0.10 to 0.60, and 0.80 at a rate of 0.03 on the 56-cluster stratum (§A.3), which suggests that small-rate intervals may be narrower than they claim. Two earlier interval constructions in this programme were found by review to cover 0.416 and 0.075 where 0.95 was claimed.

The validation instrument shares the failure mode under study. Reports were checked by a language model, which is what this paper measures a ceiling on. Studies cleared review at between three and twenty-one rounds, with the round before the last always returning substantive findings. Measurements moved on review in three studies; elsewhere the refusals concerned interpretation, where the instrument is weakest. Withholding our conclusions from one pass produced six substantive findings that six briefed passes had missed. The defect class it leaves open is the one that looks right to a language model, and we do not know what remains in it.

The reviewer is one of the measured families. `gpt-6-astra` reviewed every report in this programme and is also an auditor route whose ranking against the others the paper reports. For claims about `astra`’s relative performance the validation is a self-assessment, and those comparisons should be treated as bracketed until a model that audited nothing here has checked them.

No human rater. Every rating and adjudication in this work was made by a model, one of them the agent that designed the studies and wrote the rubrics (§A.4). A third model rating does not discharge that, and the design that asks for an outside human rater has not been run.

Budget. Reading depth K , the number of arms, and the size of every intervention population were set by cost rather than by a power calculation, and no sample size was justified in advance by a minimum-detectable-effect argument.

F. Related work in full

Critics that read code. Trained critic models write natural-language critiques of model-written code and catch more bugs than unaided humans (McAleese et al., 2024). That work evaluates on both deliberately inserted and naturally occurring bugs. On code containing natural model errors its critiques are preferred to human critiques in 63% of cases, and it surfaces errors in training data previously rated flawless. It also reports that critics hallucinate bugs, and that a human–model team hallucinates less than the model alone. The natural-defect setting and the precision cost of a louder critic are therefore

already its territory, and we claim neither. Its reported quantity is a preference or a per-critique comparison. What we add on this axis is narrow: the union of repeated independent readings as an instrument for locating a limit, ground truth supplied by an interpreter, and the false-positive rate carried beside every recall figure. Aggregating several review passes is known to help. On 1,000 verified pull requests, multi-review aggregation raises F1 by up to 43.67% (Zeng et al., 2025). The motivation is shared with our union-of- K , but there an additional model call synthesises the sampled reviews, while our union counts an instance whenever any reading flags it, so the two estimate different things.

Repeated sampling and coverage. Coverage, the fraction of problems solved by at least one of K samples, rises steeply and then flattens. This is an established regularity of repeated sampling, reported over four orders of magnitude of K and fitted by a power law (Brown et al., 2024). Our union-of- K is that quantity with a flag rule in place of a correctness check, so §B.1 re-measures a known shape, and at $K \leq 8$ our budget is small beside theirs. What is ours is the reading: the curve is taken as a measurement of a ceiling, reported beside the false-positive rate it costs, and tested against a registered flattening bar that it fails, which is why we report diminishing returns and no limit.

Self-assessment. Two lines of work bear on a model judging its own output. Self-correction without external feedback does not reliably improve reasoning and can degrade it (Huang et al., 2024). Models used as judges prefer their own generations, mostly legitimately, since stronger generators are better judges, but harmfully on the instances they themselves get wrong (Chen et al., 2025). Our same-vendor and self-audit arms are the code-audit case of that literature, and we read our null there as consistent with it. What our arms add is the false-positive rate beside the recall.

What the hidden suite decides. EvalPlus rebuilt HumanEval and MBPP with larger suites and reported defects in the original reference solutions (Liu et al., 2023); its code carries bespoke oracles for tasks where differential testing against the reference did not apply. Richter & Papadakis (2026) show that models collapse onto a single incorrect reading of an underspecified task, and name twelve MBPP tasks they identified by hand. Eleven fall inside our defect population, and our raters, who labelled them before that paper was fetched, agree on nine. On the test-generation side, the dominant error in generated tests is a wrong expected value rather than an invalid input (Xiong et al., 2023), and generated oracles tend to capture what a program does rather than what was intended (Konstantinou et al., 2024). Our claim is the auditing counterpart: where the prose does not fix the value, a recall number computed against that suite measures the benchmark as much as the auditor.

Consensus as a substitute for ground truth. Several systems validate generated tests without a reference by agreement, through self-consistency, dual execution, or multi-agent discussion for JUnit suites (Taherkhani et al., 2026; Xu et al., 2025). Both measure what survives the filter against a ground-truth implementation: 84–91% of retained tests valid in one (Taherkhani et al., 2026), 0.89–0.93 oracle correctness in the other (Xu et al., 2025). Our measurement differs in unit and in bar. It is the share of wrong tests among retained *failing test–candidate applications*, against a preregistered 2% threshold, over a frozen set of generated tests and candidates that includes candidates already failing their visible tests. Corroboration across independent generations does not reach it (§B.5), and wrong tests recur by problem. That recurrence is compatible with the self-consistent errors, the same wrong answer produced again, that degrade the detection methods Tan et al. evaluate (Tan et al., 2025). Our records hold indices, not assertions, so they do not show that the recurring errors are the same one.

Auditing scientific work. Language models are already asked to check science at several levels. On manuscripts, SPOT finds that no model exceeds 21.1% recall or 6.1% precision on errors serious enough for errata or retraction, and that repeated runs rarely rediscover the same errors (Son et al., 2025). SciCoQA asks models to find discrepancies between a paper and its code, and the best detect 46.7% of real ones (Baumgärtner & Gurevych, 2026). Closest to our setting, Luo et al. (2025) have an LLM auditor inspect AI-scientist projects for methodological pitfalls, labelled by hand, and report both true- and false-positive rates; giving the auditor the run logs and code raises detection well above reading the paper alone. scicode-lint targets methodology bugs such as leakage in scientific Python, with precision partly judged by a model (Samsonau, 2026). On data, agents that clean deliberately corrupted tables find row-level errors and miss errors that span rows (Bendinelli et al., 2025). And a deterministic verifier shows that simulation code can run and still solve the wrong equations (Song et al., 2026). SciCode, on which our code study builds, supplies scientist-written numeric tests (Tian et al., 2024), and reproduction benchmarks such as CORE-Bench ask agents to recover reported results from a paper’s code and data (Siegel et al., 2024). We found no prior measurement of an LLM auditor’s recall *and* false-positive rate against an executed oracle on scientific code, results or data, across vendors and repeated readings, set beside the deterministic checks a shipped system would run on the same artefacts. Section 4 makes that measurement.

Who audits whom. That an evaluator should not be the model under evaluation is supported by measurements of self-preference in LLM judges against verifiable ground truth (Chen et al., 2025). We do not measure self-preference. A frozen

generator’s code audited by routes that differ in model, operating point and sampling cannot separate preference from performance, so we report the same-vendor contrast as a contrast between routes, on one substrate.

Preregistration. Preregistering an evaluation protocol is beginning to appear in machine learning, both as a protocol for experiments with agents (Vaccaro, 2026) and as a defence against tuning prompts, decoding parameters or output format until a desired result appears (Thomas et al., 2026). We count our use of it as method, not contribution.

G. Reproducibility and claims

The repository at <https://github.com/dongzhaohe321418-lab/audit-ceiling-paper> carries the claims ledger (CLAIMS.md), with the claims the evidence does not support; the results report of every study the paper draws on; every registered study’s preregistration and numbered amendments; every independent review of those reports, verbatim and including refusals; and the derived records (counts, labels, hashes and intervals) that the figures and Table 1 are drawn from. A single command, `reproduce.sh`, regenerates Figures 1–4 and Table 1 from those records and checks that each is byte-identical to the copy the manuscript compiles, with no model calls, no network access and no path outside the checkout; it was run from a fresh clone with the home directory unavailable to establish that. The repository’s README maps each claim to its report, its reviews and its records. The corpora are HumanEval (MIT), HumanEval+ and MBPP+ (Apache-2.0), SciCode (Apache-2.0) and four UCI datasets (CC BY 4.0), which permit redistribution. What a replication cannot do from this repository alone is regenerate the records themselves. With the archive of raw readings, which is kept separately, they regenerate deterministically through the analysis code in the auditor harness, a separate repository named in the provenance file; without it, only by re-running the studies against model endpoints whose outputs are not deterministic. The repository is private until the paper is released.